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SCHOOL OF NATURAL AND APPLIED SCIENCES

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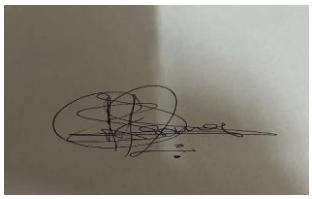
Automated Extraction and Analysis of Timing-Based Paradata Features for Survey Quality Assurance
REPORT

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Declaration

I declare that this report is my own work. It has not been submitted, in whole or in part, for any other award at this or any other institution, and every source I have drawn on is acknowledged in the reference list.

The use of artificial intelligence tools during this project is disclosed in full in Appendix A, which records what each tool was used for, what was given to it, and what I accepted, modified or rejected as a result. All evaluation data in this study is synthetic and machine generated. No respondent data, and no data of any kind belonging to a third party, was uploaded to any external tool at any point.

Signed:  Date: .....

Conventions used in this report

Precision, recall and F1 are quoted to **four decimal places whenever a single measured run or detector component is named**, in prose and in tables alike, so that values such as 0.9570 and 0.9615 stay distinguishable rather than both collapsing to 0.96. Two decimal places are used for three things only: a range spanning several runs, as in precision between 0.96 and 1.00; the proposal's success criteria of 0.80 and 0.75, which were fixed at that precision; and per-class recall, which is always accompanied by the exact counts it derives from, as in 93 of 132 at 0.70. Counts are exact throughout. Effect sizes follow the same rule as the primary metrics and are quoted to four decimal places, so that features whose effects differ only in the third decimal remain distinguishable. Runtimes are wall clock seconds for a single named run, and the run is always named. Feature identifiers of the form Q-01, S-01, V-01 and T-01 are stable across the code, the output datasets and this report.

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Abstract

Digital data collection platforms such as SurveyCTO and ODK Collect record detailed process data, commonly called paradata, alongside every interview. These audit logs hold timestamps, navigation events and validation errors that describe how an interview was conducted rather than what was answered. Survey methodology research has shown that such timing data can expose falsified interviews and speeding, yet paradata remains underused in practice. The platforms' own quality tools act field by field or interview by interview, and the raw logs differ across platforms, so turning them into a comparable set of interview-level indicators is left to each survey team. This study designed, implemented and evaluated an automated pipeline that parses SurveyCTO and ODK audit logs into a common intermediate representation, computes forty timing-based features across question, section and survey levels of granularity together with a statistical characterisation of each response time distribution, and flags anomalous interviews using interquartile range thresholds combined with transparent, configurable rules. The system was evaluated against synthetic benchmark datasets with labelled injected anomalies, including datasets generated from random seeds that were never used during development and a stress condition in which normal and anomalous behaviour deliberately overlap. Across a class imbalance sweep from five to thirty percent anomaly prevalence the detector achieved precision between 0.96 and 1.00 with full recall. On held-out stress datasets in both platform formats it achieved precision between 0.96 and 1.00 and recall between 0.84 and 0.89, exceeding the predefined success criteria of 0.80 precision and 0.75 recall. The duration-only check that represents current practice recovered 66 of 100 anomalies on the canonical benchmark and none of the 400 in the held-out stress runs, where a single navigation rule recovered between 35 and 51 percent at precision below 0.22. When the same generated interviews were written in both formats, all forty features and every detector decision were identical, so the differences between the two formats' runs reflect different sampled interviews rather than the parsers. Across the

stress runs speeding was detected in all 132 cases, disengagement in 121 of 136, and mild fabrication in 93 of 132. A diagnostic pass on those missed fabrications, run on development data only, found that response time entropy separates them from genuine interviews without overlap, so the shortfall is traced to threshold calibration rather than to any absence of signal in the features. The deliverables include a documented Python feature library, a paradata format guide, labelled validation datasets and this report.

1. Introduction

The quality of survey data underpins research, policy formulation and decision making in public health, economics and the social sciences (Groves et al., 2009). Digital data collection platforms have substantially improved the efficiency of field survey administration, and two of the most widely deployed platforms, SurveyCTO and Open Data Kit (ODK), automatically record auxiliary process data known as paradata (Couper, 1998; Kreuter, 2013). Paradata captures how an interview unfolded: when each question was opened and answered, how long the enumerator paused, whether answers were revised, and how the respondent navigated through the instrument.

These behavioural traces carry diagnostic value that conventional quality checks cannot match. Unusually fast completion may indicate rushed or fabricated responses, extended pauses may signal confusion or interruption, and atypical navigation patterns may point to satisficing behaviour, in which respondents or enumerators invest only the minimum effort required to move forward (Krosnick, 1991; Yan and Tourangeau, 2008). Despite this potential, paradata is routinely underused. Audit logs from SurveyCTO and ODK differ in structure, field naming and timing granularity, and the statistical work needed to convert millions of raw event rows into defensible quality indicators is beyond the reach of most survey teams (Kreuter, 2013).

1.1 Problem statement

Existing survey quality assurance practice relies largely on aggregate indicators, chiefly total interview duration, together with manual back-checking of a sample of interviews. These methods are poorly equipped to detect behavioural irregularities that occur during the interview, such as uniform pacing across questions of very different complexity, or timing that is compressed and unusually consistent across an interview (Bushery et al., 1999; Bredl et al.,

2012; Schwanhäuser et al., 2022). The raw material needed to detect these patterns is already collected. ODK Collect can attach an audit log of timestamped question views, jumps and form events to every submission (ODK, n.d.), and SurveyCTO records per-field timing in its text audits (SurveyCTO, n.d.-a). SurveyCTO also offers quality tools of its own, including a minimum time per field with a count of speed violations, and automated checks that warn about quality concerns in incoming data (SurveyCTO, n.d.-b). What neither platform provides is a common set of timing features computed the same way for both, combined into an interview-level score that names the signals that fired and uses thresholds suited to the skewed, non-normal distributions typical of response times (Iglewicz and Hoaglin, 1993). That gap, rather than an absence of data or of any tooling, is what this project addresses.

1.2 Aim

The aim of this project was to design and implement an automated pipeline for extracting, cataloguing and analysing timing-based paradata features from SurveyCTO and ODK audit logs in order to support scalable survey data quality assessment.

1.3 Objectives

1. To review and catalogue timing-based quality indicators identified in survey methodology literature, mapping prior findings to a structured feature taxonomy.
2. To develop format-aware parsers for SurveyCTO and ODK audit logs that produce a common intermediate representation.
3. To implement a feature extraction module computing thirty to fifty timing-based indicators across question, section and survey levels of granularity.
4. To compute statistical characterisation measures including variance, entropy, skewness and kurtosis.
5. To implement anomaly detection using interquartile range thresholding and rule-based logic, validated against precision and recall criteria.
6. To generate structured output datasets, feature documentation and a paradata format guide suitable for reuse by survey researchers.

1.4 Scope

The project processes audit logs from SurveyCTO and ODK, extracts forty timing-based and behavioural features, characterises response patterns statistically, applies interquartile range and rule-based anomaly detection, and validates the system empirically using synthetic datasets with labelled anomalies. Real-time monitoring, dashboard integration, machine learning deployment and primary data collection were excluded from scope at proposal stage and remain so.

1.5 Contributions

This work makes four contributions. The first is a format-agnostic intermediate representation for survey paradata, a six-column normalisation of SurveyCTO and ODK audit logs that decouples every downstream component from platform specifics, so that supporting an additional platform requires a new parser and no change to the feature library, the detector or the reporting layer. The second is a documented and tested feature library of forty timing-based indicators with stable identifiers spanning question, section and survey granularity together with a statistical characterisation of each interview's response time distribution, released with a coverage table that states for every feature where it is implemented, whether the pipeline exported it, and how it is tested. The third is a like-for-like baseline comparison, in which the full detector and two methods representing current practice are computed from the same raw files, the same features and the same ground truth inside a single audited script, so that the improvement over aggregate-duration checking is measured rather than asserted. The fourth is an evaluation designed to fail: held-out seeds, a stress difficulty in which normal and anomalous profiles overlap by construction, a class prevalence sweep, and a component breakdown that isolates and reports the weaker half of the author's own detector.

1.6 Structure of the report

Section 2 reviews the literature in five themes: paradata and total survey error, response time and satisficing, falsification detection, paradata in low-resource field collection, and outlier detection on skewed data. Section 3 sets out the methodology, including the common intermediate representation, the forty-feature taxonomy and the detector with its default thresholds. Section 4 describes the implementation and the defects found and fixed during development. Section 5 describes the synthetic data generator, the safeguards against overfitting, the datasets and the baselines, and section 6 reports the results. Section 7 discusses what the results support and what

they do not, section 8 states the limitations, and section 9 concludes with a reflection on the project. The appendices give the disclosure of artificial intelligence use (A), per-run confusion counts (B), the full feature validity results (C), the reproduction commands (D), the record of reference verification (E) and the map from each feature to its source (F).

2. Literature Review

This chapter reviews the work the project builds on, in five themes: paradata within the total survey error framework, response time as a measure of effort, the detection of interviewer falsification, paradata in field and low-resource collection, and outlier detection on skewed data. A sixth and shorter theme covers evaluation on synthetic and imbalanced data. Appendix E records the verification of every source cited here, including those cited only from an abstract or publisher record.

2.1 Paradata and total survey error

The Total Survey Error framework treats data quality as the accumulation of error across every stage of the survey process, from the sampling frame through measurement to post-collection processing (Groves et al., 2009). Paradata sits at the measurement stage. It does not describe what a respondent answered, but how the interview that produced the answer was conducted.

Couper (1998) set out what computer-assisted interviewing makes available for that purpose, and the description still fits the platforms used today. Once the instrument runs on a computer, he notes, "in addition to substantive data being instantly available for evaluation and review, a variety of process data are also available", so that supervisors can review the status of any case, and the number and timing of calls, while the field period is still open (Couper, 1998, p. 44). His account of keystroke files describes precisely the artefact this project consumes: keystroke files, "also called trace files or audit trails", are "automatic byproducts of many CAI software systems", and are "essentially a record of every key pressed by an interviewer (or respondent) as he/she moves through the instrument" (Couper, 1998, p. 45).

Two properties of that description shape this project. The first is that the record is a byproduct. It is produced whether or not anyone analyses it, so using it adds nothing to the cost of fieldwork, which is what makes it attractive where budgets and supervisory capacity are thin. The second is

that it records process at the level of individual actions, which means the analyst must decide what to aggregate and how. Those decisions are the feature library described in section 3.2, and they are where most of the methodological choices in this project are made. Kreuter's (2013) edited volume collects the analytic uses that follow once those decisions are taken, across a range of large survey programmes, and its organising distinction, between paradata used to manage collection while it is running and paradata used to analyse quality afterwards, runs through the rest of this chapter. This project sits on the management side: it is designed to flag interviews while the survey is still in the field and something can still be done about them.

2.2 Response time and satisficing

The reason timing carries information about quality is cognitive. Answering a survey question properly requires comprehension, retrieval, judgement and response selection, and Krosnick (1991) argues that respondents for whom this is effortful may satisfice rather than optimise. Summarising the account later, he describes such respondents as people who "shortcut the cognitive processes they execute when answering questions" (Krosnick, 1999, p. 39), and ties the behaviour to respondents who "lack the skills and motivation to answer thoughtfully" and to questions that demand difficult cognitive tasks (p. 41). Effort that is not spent does not appear in the answer, but it can appear in the clock. Yan and Tourangeau (2008) supply the other half of that argument, what determines how long an answer takes. Modelling web survey response times at the level of both question and respondent, they find that response times depend on the length and structure of the question, which largely reflects reading time, on the number and type of answer categories, on the position of the question within the questionnaire, and on respondent characteristics including age, education and experience with web surveys (Yan and Tourangeau, 2008, p. 51). They are explicit about what their study does not address: its emphasis "is not on the relation between response time and data quality per se" (p. 56). They are cited here for what drives response time, not for any claim that fast answers are poor ones; that link is what the next two studies test.

Two studies turn this into an operational threshold, and the difference between them is the one this project has to resolve. Zhang and Conrad (2014) set a speeding threshold at 300 milliseconds per word, a rough estimate of reading speed, multiplied by the number of words in the question, reasoning that a response given faster than the question could plausibly have been read cannot

rest on having read it. They find that speeding is associated with straightlining, that is with giving identical answers down a grid, and that the tendency is stronger among younger and less educated respondents. Conrad et al. (2017) use the same style of benchmark at 350 milliseconds per word, and state the principle explicitly: they define speeding "as an absolute rather than relative phenomenon: response time below a psychologically based threshold" rather than a response that is merely faster than other responses. They then show that prompting respondents who cross that threshold reduces speeding on later questions.

The alternative is a relative definition, in which an interview counts as fast because it is fast compared with the rest of the sample. This project uses both kinds of threshold, and Table 1a says which is which: absolute floors catch physically implausible speed, and median-relative thresholds catch interviews that are out of step with the instrument and population they belong to. Two limits of this literature should be stated. It studies respondents completing web surveys themselves, not enumerators administering interviews in the field, and its thresholds are set per question in a known instrument, whereas an audit log arrives without the questionnaire that produced it. Neither difference removes the underlying argument that unusually fast answers indicate unspent effort, but both mean the constants cannot simply be transplanted.

2.3 Detecting interviewer falsification

Where satisficing concerns effort, falsification concerns fabrication: an interview recorded as conducted that was not conducted, or not conducted as recorded. The timing literature on falsification begins with Bushery et al. (1999), who propose using the date and time stamps that computer-assisted instruments record to identify suspect interviewers. Their reasoning is stated plainly: "We hypothesize that fabricating an interview would be faster than a comparable real interview", and so "Time stamps can reveal extremely short interviews", with interviewers producing significantly more short interviews than the norm selected for a focused reinterview check (Bushery et al., 1999, p. 317). They apply statistical process control charts to weekly interviewer output, and they set three criteria for such screening that this project is also bound by: it must identify interviewers who deviate significantly from the norm, it must not identify so many that the review process becomes unworkable, and it must return a higher proportion of confirmed falsification than a random reinterview would.

Bredl et al. (2012) place timing within a wider set of indicators. Reviewing the approach, they note that interviewers who "need a remarkably long or short time to complete the entire questionnaire or certain modules" are candidates for suspicion (Bredl et al., 2012, p. 2), and their own method combines several indicators, including the distribution of leading digits in monetary variables, in a cluster analysis that separates suspicious from unsuspicious interviewers. Schwanhäuser et al. (2022) extend that comparison systematically. Working with the IAB-BAMF-SOEP Survey of Refugees in Germany, in which three falsifiers had been confirmed through routine quality control, they assemble 32 indicators, including both absolute and relative interview duration, and compare cluster-analytic methods against a meta-indicator that standardises and sums the indicators for each interviewer. The best-performing approaches identified the known falsifiers with very high accuracy and without false negatives, and their practical conclusion is that because individual falsifiers score inconsistently on any single indicator, combining as many indicators as possible is the sounder strategy. They also report the pattern that recurs throughout this literature: falsified interviews show lower variance, within and between interviews, than genuine ones.

The closest prior work to this project is Birnbaum et al. (2013), who instrument ODK Collect, the same platform this project parses, to record low-level behavioural logs with millisecond-precision timestamps, and apply supervised classification to features extracted from them. Their features are organised much as the ones here are: prompt-level features for each question, instance-level aggregations such as total time and total, median and minimum answer time, and interviewer-normalised versions of both, expressed as the number of standard deviations between an interview's value and that interviewer's own mean (Birnbaum et al., 2013, p. 2913). Their classifiers identify fabrication "with an accuracy of up to 96%", and still reach 86 percent when interviewers know the method is in use, understand roughly how it works, and are paid in a way that rewards evading it (Birnbaum et al., 2013, p. 2911). They also show that the behavioural data is what carries the result: without it, accuracy falls to little better than chance.

Three things follow for this project. The timing signature of fabrication is consistent across this literature: faster, more compressed, less variable. The consensus method is to combine many indicators rather than to trust one. And the strongest results come from supervised classification,

which requires labelled fabrication to train on. This project departs on the last point, using transparent univariate rules that need no labels, and section 6.3 reports what that costs.

2.4 Paradata in field and low-resource survey settings

The literature above is drawn largely from well-resourced survey programmes, and the deployment case for this project rests on settings that look different. The platform this project targets was built for those settings: Hartung et al. (2010) present ODK as an "open-source suite of tools designed to build information services for developing regions" (p. 1), comprising Collect for mobile data entry, Aggregate for server-side storage, and Build and Voice alongside them, and describe four deployments that exercise the design. Both platforms now record process data as a standard feature rather than as an add-on. ODK Collect can attach an audit log to every submission, recording each event, the node it concerns, and start and end times in epoch milliseconds (ODK, n.d.). SurveyCTO records per-field timing in its text audits, and offers quality tools of its own, including a minimum time per field with a count of speed violations, and automated checks that warn about quality concerns in incoming data (SurveyCTO, n.d.-a, n.d.-b).

What is scarce in these settings is not the data but the capacity to act on it. Hoogeveen and Pape (2020) collect World Bank experience of collecting data in fragile and low-capacity states, with the explicit aim of shifting the question from "there are no data" to "how do we go about collecting data here?" (Hoogeveen and Pape, 2020, foreword). Supervisory capacity is thin, back-checking a meaningful share of interviews is often impractical, and collection runs on the same mobile platforms this project targets. Finn and Ranchhod (2017) supply the counterfactual that makes the problem concrete: fabrication in South Africa's National Income Dynamics Study, affecting about 7 percent of the sample, was detected while fieldwork was still under way and the affected interviews were re-conducted, and they find that it would not have affected their univariate and cross-sectional estimates meaningfully, but would have led to substantially different conclusions once panel estimators were used (Finn and Ranchhod, 2017, p. 129). Fabrication in field surveys is therefore real, occurs at measurable rates, and is detectable while collection is still in progress, which is the window a paradata pipeline operates in; and its consequences are uneven enough that detecting it is a better strategy than reasoning about whether a particular analysis would survive it.

The most directly comparable deployment is Goel, Abraham and Lahoti (2022), who used paradata during the India Working Survey to identify enumerators whose field practice deviated from that of their peers, and fed the finding back to them through their supervisors while collection was still running. Two features of their design bear on this project. The first is dynamic benchmarking: they compare each enumerator against a group facing similar field conditions rather than against a fixed constant, which is the reasoning behind the median-relative thresholds in Table 1a. The second is a caution this project adopts in section 7. A flag, they write, "should not be construed as conclusive evidence of wrongdoing", because "while it correctly identifies deviant behavior, it does not go into the reasons for it" (Goel et al., 2022, p. 11).

Three implications follow for the design here, and each is honoured in the implementation. Detection must be explainable to a field supervisor, because a flagged interview leads to a conversation with a named enumerator rather than to a modelling decision, which is why the anomaly score is a count of named flags rather than a model output. Thresholds must be expressible relative to the sample rather than as absolute constants wherever the instrument and the population determine what is normal, which is why the thresholds most likely to depend on context are median-relative and every threshold is configurable (Table 1a). And review capacity is scarce, which is why the precision behaviour at low anomaly prevalence reported in section 6.4 is treated as an operational result rather than as an artefact of the metric.

2.5 Outlier detection on skewed data

The detector's statistical layer rests on Tukey's boxplot fences, which mark as potential outliers any value outside the interval from the first quartile minus one and a half interquartile ranges to the third quartile plus the same margin. The construction uses only order statistics, so computing it requires no distributional assumption, and it is resistant to the influence of the very values it is meant to identify.

Its flagging rate, however, is not assumption-free, and Hubert and Vandervieren (2008) quantify the gap. For a normal distribution the expected proportion of observations outside the fence is only 0.7 percent, but for a chi-squared distribution with one degree of freedom 7.56 percent of regular observations can be expected to exceed the upper fence, and for a lognormal distribution 7.76 percent; in their worked example of intervals between coal mining disasters, 6.84 percent of observations fall beyond the upper whisker without being outliers in any meaningful sense. Their

conclusion is blunt: because textbooks and software rarely distinguish potential outliers from real ones, "many practitioners nowadays incorrectly use the boxplot as a tool for outlier detection" (Hubert and Vandervieren, 2008, §1). The distribution of response times is, in Yan and Tourangeau's words, "not normal, but asymmetrical and positively skewed" (2008, p. 52), which places this project squarely in the case Hubert and Vandervieren warn about, and it is why the statistical layer is read here alongside the rules rather than on its own.

Their review of earlier corrections bears directly on section 7. Some authors adjusted the fence for skewness by replacing the interquartile range with the lower and upper semi-interquartile ranges, the distances from the median to the first and third quartiles, giving a fence from the first quartile minus three lower semi-interquartile ranges to the third quartile plus three upper ones (Kimber, 1990; Aucremanne et al., 2004, as cited in Hubert and Vandervieren, 2008, §1). The entropy fence trialled in section 7 is the lower-side analogue of exactly that construction, with the same constant, so it is not an invention of this project. Hubert and Vandervieren also judge the semi-interquartile fence to adjust insufficiently for skewness, which is a documented limitation of the correction and is reported as one. Their own proposal replaces it with a fence scaled by the medcouple, a robust measure of skewness, chosen so that the skewness estimate is itself not distorted by the outliers being sought.

The broader anomaly detection literature supplies the vocabulary for what the detector does and does not attempt. Chandola et al. (2009) distinguish data instances described by a single attribute, the univariate case, from those described by several, the multivariate case (§2.1), and note that unsupervised techniques, which require no training data and are therefore the most widely applicable, "make the implicit assumption that normal instances are far more frequent than anomalies in the test data. If this assumption is not true then such techniques suffer from high false alarm rate" (§3). This project violates that assumption deliberately in the prevalence sweep, and the consequence observed in sections 6.3 and 6.6 runs in the opposite direction to the one they describe: because the fences are estimated from the contaminated sample, rising contamination moves the quartiles outwards and costs recall rather than precision. Hodge and Austin (2004) survey the same field from the statistical side and confirm that many classical outlier methods are single-dimensional, and that boxplots are among the tools used on univariate data. Univariate methods of this kind are transparent and interpretable, properties valued in

survey quality workflows where a flagged interview must be explainable to a field supervisor, but they cannot detect anomalies visible only in combinations of features. That trade-off is accepted explicitly in the present design and revisited in the results.

2.6 Evaluation on synthetic and imbalanced data

Evaluation metrics computed on data with injected anomalies are sensitive to class prevalence, so a fixed anomaly ratio can misrepresent operational performance; varying the ratio, as the class imbalance sweep in section 6.4 does, is the standard mitigation (Japkowicz and Stephen, 2002). The generation of the anomalies themselves rests on weaker ground. The first submission cited Chawla et al. (2002) for the claim that injecting known anomaly types into generated data is an established way to benchmark detection systems. That paper describes SMOTE, a method for oversampling a minority class in training data, and supports no such claim, so it has been withdrawn. No source has been substituted, because none was found that establishes the practice for this setting. The generator and its injected profiles are therefore presented here as a design choice of this project, with the construct-validity risks that follow set out in sections 5.2 and 8.

The nearest precedent in the falsification literature runs the other way, and is instructive about the cost. Birnbaum et al. (2013) obtained labelled fabrication experimentally, by having interviewers conduct real interviews and also fabricate them under controlled conditions, so their labels describe human behaviour rather than a generator's assumptions. That is the stronger design, and the limitation it exposes in this one is stated in section 8: a synthetic evaluation measures a detector against a documented threat model, not against the full range of behaviour a real enumerator might produce.

3. Methodology

The system is a multi-stage pipeline that proceeds from raw log ingestion through parsing, feature extraction, statistical characterisation and anomaly detection to structured output generation. Development followed an iterative, test-driven approach in Python 3.10 and later versions, using Pandas and NumPy for data processing, SciPy for statistical functions, Matplotlib and Seaborn for visualisation, and Pytest for automated testing (McKinney, 2017).

3.1 Common intermediate representation

SurveyCTO and ODK audit logs differ in structure, field naming and timestamp encoding. ODK Collect's form audit log is an event log: each row records an event such as form start, question, jump or end screen, the node it concerns, and start and end times in epoch milliseconds (ODK, n.d.). SurveyCTO's text audit is by default a per-field summary of total duration and first-appeared time in seconds, and becomes an event log with device time, form time, event and duration in milliseconds when the eventlog appearance is set (SurveyCTO, n.d.-a). To decouple analysis from these differences, every parser converts its input into a common intermediate representation, a standardised table with six columns: interview identifier, field name, event type, start time in milliseconds, end time in milliseconds, and duration in milliseconds. All downstream components operate exclusively on this representation, so support for an additional platform requires only a new parser and no change to the feature library.

The parser layer is a class hierarchy. An abstract base parser defines the intermediate schema and validates that every parser output conforms to it. Concrete subclasses for SurveyCTO and ODK handle ingestion, event name normalisation and timestamp conversion.

The two parsers were developed against the synthetic exports described in the paradata format guide, and a check against the platforms' documentation during revision showed that those layouts are not the native ones described above. The synthetic SurveyCTO files are event rows carrying a node, an event and start and end times in epoch milliseconds, which is closer to ODK's native audit log than to either SurveyCTO format. The synthetic ODK files use instance and name columns with ISO 8601 timestamp strings, a layout ODK does not produce. A test log written to ODK's documented layout was used to check what the pipeline does with one, and two gaps it exposed have since been closed. Automatic format detection now recognises a native ODK audit log by its event vocabulary, since the column names alone do not distinguish it from a SurveyCTO export, and routes it to the ODK parser. ODK's remaining form-level events, form save, form exit, form resume and form finalize, are normalised alongside form start and end screen and excluded from the section statistics, which previously they were not: carrying no node, they formed a pseudo-section whose gaps distorted every pause measure, the same distortion the administrative-event fix in section 4 was meant to prevent. A test in the suite covers both behaviours. What remains unsupported is SurveyCTO's native text audit, in either its

default per-field form or its eventlog form, which neither parser reads. No native export from either platform was used in the evaluation. Section 8 records this as a limitation of the claims about platform support.

Each parser maps its native vocabulary onto a shared set of normalised event types so that a form opening, a question interaction and a navigation jump mean the same thing downstream regardless of origin.

Timestamp handling deserves particular care because it is where real logs most often deviate from documentation. The SurveyCTO parser accepts epoch milliseconds as its canonical input but includes a guard for exports that record epoch seconds, detected from the magnitude of the median value and corrected by rescaling. The ODK parser converts ISO 8601 strings, including timezone offsets, into epoch milliseconds. Both parsers tolerate missing field names, unparseable timestamps and negative durations by marking the affected rows rather than silently accepting or discarding them, so that data problems surface in the quality flags rather than corrupting features. A controller above the parsers performs automatic format detection from column structure and timestamp shape, and can split hybrid files in which rows from both platforms have been concatenated into a single export, a situation that occurs in practice when field teams merge files before upload.

3.2 Feature taxonomy

Forty features were catalogued at proposal stage from the literature reviewed above and all forty are implemented. They are organised in four groups with stable identifiers used consistently in the code, the documentation and the output datasets.

Table 1. Feature taxonomy, forty catalogued features in four groups.

Group	Identifiers	Content
Question level	Q-01 to Q-10	Response time, time to first interaction, edit count, revisit flag and count, completion ratio, idle time, event count, constraint triggers, first-to-last interaction span
Section level	S-01 to S-10	Mean section pace, section duration, within-section variance, pace ratio to survey mean, pause counts above

Group	Identifiers	Content
		five and thirty seconds, mean and maximum pause, back-navigation count, completion rate
Survey level	V-01 to V-10	Total duration, early and late pace, pace trajectory ratio, back-navigation rate and total, start hour, day of week, duration relative to sample median, consecutive no-back-navigation flag
Statistical	T-01 to T-10	Variance, interquartile range, Shannon entropy, skewness, excess kurtosis, mean, median, coefficient of variation, minimum and maximum of the response time distribution

Several features are proxies or coincide by construction, and the feature validity results in section 6.2 cannot be read correctly without knowing which. The event count per field (Q-08) stands in for keystroke count, because neither platform logs individual keystrokes, and the field completion ratio (Q-06) is derived from whether measurable time was spent on the field, because the audit log does not record answer content. Three further features are proxies of the same kind. The edit count (Q-03) is the number of events on a field minus one, so on these logs it carries the same information as Q-08, and the two have identical effect sizes in Appendix C. Time to first interaction (Q-02) uses separate field start and end markers when a log has them and otherwise falls back to the duration of the first event, which on logs with one event per question visit is close to the response time itself. Section completion (S-10) is measured against the most complete section observed in the same interview, because no questionnaire schema is available at run time.

Three implementation details also shape what the features can show. The section back-navigation count (S-09) matches events named back or backward, while the survey-level navigation features V-05, V-06 and V-10 also match jump, which is the event name ODK uses and the only navigation event the synthetic generator emits; S-09 is therefore zero throughout the synthetic corpus. The section pace ratio (S-04) is computed per section as section pace over the interview's mean pace and then averaged over sections, which equals one by construction whenever sections contain the same number of questions, as they do in every benchmark here.

And the interview-level values of Q-01, T-06 and S-01 are all means of the same per-question response times, so they coincide when sections are of equal size, which is why they share one row of effect sizes in Table 2. All of these properties are documented in the feature guide so that users do not over-interpret the features.

Administrative events require care. Real audit logs open and close with form-level rows whose field identifier is blank. These rows contribute to total interview duration but are excluded from question-level and section-level statistics, because treating the gap between form start and form end as a pause would swamp every within-section measure. This exclusion was added after exactly that distortion was observed during validation.

3.3 Key statistical formulations

Three of the statistical features carry most of the detection weight and merit explicit definition. Shannon entropy of the response time distribution measures the unpredictability of an interview's rhythm. Response times are binned into one-second intervals, the proportion of questions falling in each occupied bin is computed, and entropy is the negative sum of each proportion multiplied by its base-two logarithm. A fabricator who paces every question similarly concentrates the distribution in one or two bins and drives entropy towards zero, while a genuine interview spreads across many bins. The one-second bin width is fixed so that identical behaviour yields identical entropy in both platform formats regardless of timestamp precision.

The coefficient of variation, the standard deviation of response times divided by their mean, provides a scale-free measure of pacing uniformity. Because it is normalised by the mean, it treats a uniformly slow fabricator and a uniformly fast one identically, catching mechanical pacing at any speed. Values below one tenth indicate pacing more uniform than human respondents produce across questions of varying complexity.

Outlier fences follow Tukey's construction. For a given feature the interquartile range is the distance between the first and third quartiles, and values falling more than one and a half times that distance outside either quartile are flagged. The construction uses only order statistics, so it makes no distributional assumption and is itself insensitive to the outliers it identifies, which is essential for right-skewed response time data where a z-score approach would let extreme values distort their own detection threshold (Iglewicz and Hoaglin, 1993).

3.4 Anomaly detection

An anomaly is defined at the interview level as statistically significant deviation from expected timing behaviour. Detection combines two complementary mechanisms. Statistical flagging applies Tukey fences to four features: total duration, mean response time, response time entropy and response time variance. For each, interviews below the first quartile minus one and a half times the interquartile range, or above the third quartile plus the same margin, receive a flag. Rule-based flagging applies domain-derived thresholds covering implausibly short or long sessions, low mean response time relative to the sample median, low entropy, uniform pacing measured by the coefficient of variation, compressed interquartile range, low within-section variance, absent back-navigation in compressed-timing contexts, elevated back-navigation rates, and mean inter-question pauses well above the sample median. The thresholds most likely to depend on the instrument or population are expressed relative to the sample median; the rest are absolute. Every threshold, including the fence multiplier of 1.5, is a configuration parameter, with the defaults shown in Table 1a, and a configuration file can override any of them.

Table 1a. Detector flags and default thresholds. Median-relative thresholds use the median of the positive values in the batch being scored. Conditional flags fire only when at least one compression flag (low entropy, uniform timing, low response-time IQR or low within-section variance) has already fired.

Flag	Feature	Fires when, at the default	Form
Too fast	V-01 total duration	below 6,000 ms	absolute
Too slow	V-01 total duration	above 3,600,000 ms (one hour)	absolute
Short relative duration	V-09	below 0.5	median-relative
Low mean response time	T-06	below 0.5 × median	median-relative
Low entropy	T-03	below 0.5 bits	absolute
Uniform timing	T-01, T-08	variance below 10 ms²,	absolute, scale-free

Flag	Feature	Fires when, at the default	Form
		or coefficient of variation below 0.1	
Low response-time IQR	T-02	at or below 0.5 × median	median-relative, conditional
Low within-section variance	S-03	at or below 0.25 × median	median-relative, conditional
Absent back-navigation	V-10, V-06	V-10 = 1 or V-06 = 0	conditional
Moderate back-navigation	V-05	0.25 or above	absolute rate
High back-navigation	V-05	0.5 or above; the moderate flag also fires	absolute rate
Long pause	S-08	at or above 2.0 × median	median-relative
Elevated pause	S-07	at or above 2.5 × median	median-relative
Pause rule	S-06, S-08	two or more pauses over 30 s and a maximum pause of at least 280,000 ms	absolute
Invalid ratio (data quality, also scored)	V-04	missing	not applicable
Interquartile fences, four flags jointly capped at one point	V-01, T-06, T-03, T-01	below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR	sample quartiles
Category: suspicious	anomaly score	2 or above	fixed default
Category: anomalous	anomaly score	5 or above	fixed default

Each interview receives an anomaly score equal to the number of active flags, with the four interquartile fences contributing at most one point jointly. The cap exists because the four fenced quantities, total duration, mean response time, entropy and variance, are driven by the same underlying pacing: an interview that is compressed in one is usually compressed in the others, so counting each fence separately would let a single behaviour score up to four points. Interviews scoring at or above two are categorised as suspicious and at or above five as anomalous. The cut-off of two means that no single rule and no single fence can put an interview under review on its own; in the canonical suite every genuine interview scored at most one. The cut-off of five requires agreement across several families of evidence, typically duration, distributional shape and navigation together; in the canonical suite fabricated interviews scored five or six, speeding interviews three or four and disengaged interviews two or three. Both cut-offs were therefore set on synthetic data from the same generator later used for evaluation, which is the construct-validity risk discussed in section 5.2. They are defaults to be recalibrated on field data, for which `scripts/threshold_recalibration.py` is provided, not estimates of an optimal threshold. A separate set of data quality flags, covering impossible values such as negative durations or a maximum below a minimum, routes technically corrupt records into their own category so that device faults are not misread as behavioural anomalies. The score is a transparent count: every flagged interview can be explained by listing its active flags, which is a requirement for use in field supervision.

3.5 Computational considerations

The pipeline was designed for datasets ranging from pilot studies of a few dozen interviews to national surveys of tens of thousands. Feature extraction uses vectorised Pandas operations whose cost grows approximately linearly with the number of audit events per interview, and anomaly detection is dominated by quantile computation over interviews. Files are processed one at a time, but the batch path keeps the question-level and section-level tables for every interview until they are written out, so memory grows linearly with the number of interviews; section 6.7 reports the measured peak.

3.6 Ethical considerations and AI usage disclosure

The system operates on paradata rather than on survey responses, which substantially reduces privacy risk. Paradata is not free of ethical weight, however. Timing records describe the conduct

of a named enumerator, and a flag raised by this system is an accusation about a person's work, which is why the design commits to explainability: a supervisor acting on a flag can see exactly which named rules fired and check them against the interview. The severity bands exist for the same reason, to separate an interview that warrants a question from one that warrants investigation.

All evaluation data in this study is synthetic and machine generated from recorded random seeds, so no respondent information of any kind was processed, and no data belonging to any third party was uploaded to any external tool at any point in the project.

The use of artificial intelligence tools during development is disclosed in full in **Appendix A**, which records each episode of use, the tool and version where recoverable, what was provided to it, the category of any data involved, what I accepted, modified or rejected as a result, the tests attached to significant outputs, and a reflection on the limitations of the disclosure itself. The same log is maintained in the repository at `docs/ai_usage_log.md`.

4. Implementation

The pipeline is organised as five decoupled layers coordinated by a controller. Figure 1 shows the arrangement and the point at which platform differences stop propagating.



Figure 1. Five-layer pipeline architecture. Ingestion accepts audit log files, parsing normalises them into the common intermediate representation, extraction computes the four feature groups, validation applies fences and rules to produce the anomaly score, and reporting writes the datasets and evaluation artefacts. The intermediate representation is the only contract between layers 2 and 3, so a new platform requires a new parser and no change anywhere downstream.

The ingestion layer accepts one or more CSV files through a command line interface with an optional interactive mode. The parser layer produces the common intermediate representation described above. The extraction layer contains four extractor classes, one per feature group, each of which consumes the intermediate representation, or in the case of the statistical extractor the question-level output, and returns its features independently of the others. The validation layer applies the anomaly detector to the combined interview-level feature table, and the reporting

layer writes the feature dataset, a per-interview anomaly report, granular question-level and section-level tables, and evaluation artefacts when ground truth labels are supplied.

Features that require sample context are computed at batch level. Duration relative to the sample median, for example, cannot be known for a single interview in isolation, so the controller computes it once all interviews in a batch have been extracted. The evaluation path merges predictions with ground truth labels through a strict one-to-one join that fails loudly on duplicate or missing interview identifiers, and the detector never sees the labels; this separation prevents any leakage between detection and evaluation.

Every detector threshold lives in a configuration object validated at load time, with defaults matching the values justified in the methodology. A configuration file can be supplied on the command line, and a recalibration script derives a configuration from a labelled dataset by sweeping the score threshold and selecting the value that best satisfies the precision and recall targets. Configurations record their provenance, including the datasets used and the generation timestamp, so that any reported result can be traced to the exact settings that produced it. Reproducibility extends to the data: every synthetic dataset records its random seed, size, anomaly ratio and difficulty in an accompanying summary file, and regenerating any benchmark is a single documented command.

The automated test suite contains forty-three tests covering the parsers, each feature extractor, the detector, the configuration validation and the end-to-end pipeline, and the suite passes in full at the time of writing. During a line-by-line review of the codebase two defects were found and fixed: an upstream data integrity flag was being overwritten by the detector, and missing field names in ODK logs were being converted to the literal string nan instead of the intended placeholder. A third defect surfaced during evaluation, when newly added form-level events were found to be inflating section pause statistics, and was fixed by excluding administrative events from section aggregation. A final pre-submission audit in a clean environment surfaced two further defects: the conversion from ISO 8601 timestamps to epoch milliseconds assumed the nanosecond datetime resolution of earlier pandas versions and silently produced values a thousand times too small under pandas 3, and the question-level response time was computed as the first-to-last event span, duplicating a separate feature, rather than as the documented sum of event durations. Both were corrected.

Each of the five fixes now has a regression test: the preserved integrity flag (`test_detector_preserves_upstream_data_quality_flag`), the ODK placeholder (`test_odk_parser_missing_field_names_use_placeholder`), the exclusion of administrative events from section statistics (`test_section_features_exclude_administrative_form_events`), the timestamp conversion (`test_odk_parser_iso_timestamps_convert_to_epoch_ms`) and the response time definition (`test_question_features_preserve_field_level_behavior`). The first three were added during revision. In the version first submitted, the ODK placeholder and administrative-event fixes had no test, and the statement that all five fixes were covered was wrong. Three further tests were added at the same time: one confirming that the fence multiplier is configurable, one confirming that a single generated history parses identically from both formats (section 6.5), and one confirming that a log in ODK's documented native layout is detected and its form-level events excluded from the section statistics (section 3.1).

5. Evaluation Methodology

5.1 Synthetic data generation

Evaluation uses synthetic audit logs with labelled injected anomalies, because no labelled field data was available. Section 2.6 sets out why this is presented as a design choice of the project rather than as established practice. The generator writes structured logs in the two synthetic formats described in section 3.1: interviews open with a form-start row and close with a form-end row carrying blank field identifiers, questions belong to five named sections of four questions each, genuine respondents occasionally trigger validation constraints, and navigation is written as a jump row after randomly chosen questions. Table 1b gives every parameter. Response times for genuine interviews follow a log-normal distribution, a right-skewed form chosen to match the shape response time data is reported to take (section 2.5); the parameters are a design choice rather than an estimate from field data, and are scaled by a fixed per-question complexity profile so that short factual items complete quickly and recall-heavy items complete slowly. Back-navigation counts follow a Poisson distribution.

Three anomaly profiles are injected. Fabricated interviews compress the response time distribution, ignore question complexity and suppress back-navigation, which is the signature the

falsification literature reports (section 2.3). Speeding interviews scale response times below plausible completion thresholds, following the reading-speed reasoning of Zhang and Conrad (2014) and Conrad et al. (2017), though without their per-question word counts. The disengaged profile, which combines slow uniform pacing with elevated pauses and erratic back-navigation, has no comparable source in the literature reviewed here; it is a design assumption of this project, and section 8 records it as one. A fourth profile injects degenerate records, such as single rows with zero duration, to exercise the data quality path. The generator has a canonical difficulty, in which profiles are cleanly separated, and a stress difficulty, in which some genuine respondents are fast with little navigation and some fabricated interviews are only mildly compressed, so that the profiles overlap and detection is genuinely difficult.

Table 1b. Synthetic generator parameters, from `tests/test_adversarial.py`. Every interview has twenty questions in five sections of four. N(μ, σ) is a normal draw, LN(m, s) a log-normal draw with median m and log-scale s, and U[a, b) a uniform integer draw; floors are applied after drawing. Times are in milliseconds and are drawn per question. Genuine response times are multiplied by a fixed complexity weight between 0.50 and 1.40 per question; the anomalous profiles ignore the weights.

Profile	Difficulty	Response time	Back-navigation (jump) events	Pause after each question
Genuine	canonical	LN(8,000, 0.5)	Poisson(2)	U[150, 600)
Genuine	stress	one of four sub-profiles with equal probability: typical LN(8,000, 0.5) with Poisson(2); fast LN(4,500, 0.35) with Poisson(1); low-navigation LN(7,500, 0.45)	as listed	U[100, 900)

Profile	Difficulty	Response time	Back-navigation (jump) events	Pause after each question
		with none; variable LN(8,500, 0.75) with Poisson(2)		
Fabricated	canonical	N(1,200, 80), floor 500	none	U[50, 200)
Fabricated	stress	with probability 0.45, N(1,800, 180), floor 700, with no navigation; otherwise N(3,800, 650), floor 1,200	0 or 1 in the second variant	U[60, 350)
Speeding	canonical	U[200, 800)	none	U[30, 150)
Speeding	stress	U[1,200, 3,600)	0 or 1	U[40, 250)
Disengaged	canonical	N(6,000, 200), floor 4,000	6 to 10	U[400, 1,200)
Disengaged	stress	LN(6,500, 0.28), floor 3,000	4 to 8	U[300, 2,500), and with probability 0.35 one pause of U[30,000, 90,000)
Data quality	canonical suite only	a single row of zero duration	none	none

Genuine interviews also receive Poisson(0.6) constraint errors lasting U[800, 4,000) ms. Each interview opens with a form-start row lasting U[400, 2,500) ms and closes with a form-end row lasting U[200, 1,500) ms, each jump row lasts 300 ms, and benchmark interviews start ten

minutes apart. Anomalous interviews are split equally between fabricated and speeding, with the remainder disengaged, so 100 anomalies become 33, 33 and 34.

5.2 Design against overfitting

A detector evaluated only on data resembling its calibration data will report flattering results. Three safeguards address this. First, detector settings that had been tuned to earlier generator behaviour were removed or generalised before the final evaluation round: a coupling between two flags that matched the old generator was deleted, fixed pause constants became configuration parameters, and the disengagement rule was reformulated relative to the sample median. Second, all reported evaluation runs use random seeds that were never used during development or calibration. Third, evaluation includes the stress difficulty and a class imbalance sweep in which anomaly prevalence varies from five to thirty percent, following the recommendation of Japkowicz and Stephen (2002).

One risk these safeguards cannot remove is that the generator and the detector share an author. The anomaly profiles in Table 1b encode my reading of what fabrication, speeding and disengagement look like, and the rules in Table 1a encode the same reading, so a high score measures agreement between two expressions of one set of assumptions as much as it measures the detector's ability to find anomalies in field data. Held-out seeds guard against fitting to particular random draws, not against this shared construct, and the stress condition narrows the gap only by making the profiles overlap. The results should therefore be read as evidence that the detector recognises the threat model the generator documents. Section 8 returns to what this implies.

5.3 Datasets

Four kinds of dataset were used. A canonical suite of twenty-four interviews, twelve per platform format, pairs each anomaly profile with known expected outcomes and serves as a fixed regression test. Benchmark datasets of one hundred and twenty to ten thousand interviews at twenty percent anomaly prevalence exercise scale in both formats. The imbalance sweep holds size at five hundred interviews and varies prevalence across five, ten, twenty and thirty percent. The stress datasets, five hundred interviews each from two held-out seeds and rendered in both formats, provide the overlapping-profile condition. For the platform comparison in section 6.5,

the canonical n500 benchmark and both stress datasets were also regenerated with a single history written into both formats. Ground truth for every dataset records each interview's profile and binary anomaly status, and the evaluation join verifies that predictions cover the ground truth exactly.

5.4 Metrics and baselines

Primary metrics are precision, the proportion of flagged interviews that are truly anomalous, and recall, the proportion of truly anomalous interviews that are flagged, with the F1 score as their harmonic mean. Success criteria were fixed at proposal stage: precision at or above 0.80 and recall at or above 0.75. Feature-level validity is assessed with Mann-Whitney U tests comparing each feature between normal and anomalous interviews, with effect sizes reported. The full detector is compared against two baselines representing current practice: a duration-only method that flags interviews solely on total duration outside the Tukey fences, and a single-rule method, V-10, that flags every interview whose log contains a run of twenty or more consecutive events with no back or jump event. The first submission described this baseline as flagging interviews with zero back-navigation. That was wrong: an interview with one early jump followed by twenty uninterrupted events is also flagged, and zero back-navigation is a different feature, V-06 = 0.

6. Results

6.1 Canonical suite

On the canonical suite of twenty-four interviews, twenty were assigned their expected category. Genuine interviews scored one flag each and were categorised normal, fabricated interviews scored five or six and were categorised anomalous, disengaged interviews scored two or three and were categorised suspicious as expected, and the six degenerate records were routed to the data quality category rather than scored behaviourally. The four speeding interviews, two per format, did not reach their expected category: the generator labels speeding as anomalous, but they scored three or four and were categorised suspicious. They were still flagged, so binary precision and recall on the suite are both 1.0000, but the severity band understates speeding in this suite. The first submission's statement that every interview was assigned its expected category was wrong.

6.2 Feature validity

Each of the forty catalogued features was tested for its ability to separate normal from anomalous interviews on the n500 SurveyCTO benchmark, which contains 400 normal and 100 anomalous interviews. The test is Mann-Whitney U, chosen because response time distributions are strongly right-skewed and a rank-based test makes no distributional assumption (Iglewicz and Hoaglin, 1993). Because forty features are tested against the same data, p-values are adjusted for multiple comparisons using the Bonferroni correction at a family-wise alpha of 0.05. Cohen's d is reported as the primary effect size, with a rank-biserial correlation derived from the Mann-Whitney statistic alongside it (Cohen, 1988; Kerby, 2014), the latter for the simple difference formula that expresses the rank-biserial correlation as the difference between the proportions of favourable and unfavourable pairs. Bonferroni was preferred to the Benjamini-Hochberg procedure (Benjamini and Hochberg, 1995) because the question asked of each feature is whether it separates the groups at all, and a family-wise guarantee is the stricter answer to that question. The choice does not change the result: the ten features that are not significant have raw p-values of 0.59 or above, or undefined ones, so the Benjamini-Hochberg procedure would select the same thirty.

Thirty of the forty features are significant after adjustment. The largest effects are concentrated in the statistical and timing families.

Table 2. Ten largest feature effects, n500 SurveyCTO benchmark, Bonferroni-adjusted.

Feature	Cohen's d	Rank-biserial	Adjusted p
T-03 response time entropy	-10.5664	1.0000	1.6e-52
Q-10 first to last interaction duration	-4.1410	0.9983	3.0e-52
T-06 response time mean	-4.1370	0.9983	3.0e-52
S-01 mean section pace	-4.1370	0.9983	3.0e-52
Q-01 response time	-4.1370	0.9983	3.0e-52
Q-02 time to first interaction	-4.0951	0.9983	3.0e-52
S-02 section duration	-3.8040	0.9942	8.2e-52
V-01 total interview duration	-3.7853	0.9930	1.1e-51

Feature	Cohen's d	Rank-biserial	Adjusted p
V-09 session length vs median	-3.7853	0.9930	1.1e-51
T-02 response time interquartile range	-3.5061	1.0000	2.0e-52

Response time entropy is the outlier among outliers, at a Cohen's d of -10.5664 and an adjusted p of 1.6e-52, with a rank-biserial correlation of exactly 1.0000, meaning the two groups do not overlap at all on this feature in this dataset. That result is what makes the fabrication diagnosis in section 6.6 possible, and it is also what makes the shortfall reported there a calibration failure rather than a signal failure: the separation is present in the feature and the detector does not fully act on it.

The same test on the ODK rendering of the n500 benchmark (`output/feature_validation_odk/`) gives the same thirty significant features and the same ten largest effects in the same order up to ties, with entropy first at a Cohen's d of -10.7579. The ranking does not depend on the format.

Ten features do not discriminate. Two of them, V-07 start hour and V-08 day of week, are contextual rather than behavioural and would not be expected to separate the groups. The other eight carry no signal for reasons that lie in the implementation as well as the generator; the first submission attributed them to the generator alone. S-04 averages to one by construction when sections are of equal size (section 3.2). S-09 matches only back and backward events, and the generator emits jump. S-05 and S-06 count pauses over five and thirty seconds, and in the canonical corpus every generated pause is under 1.2 seconds. Q-04 and Q-05 count returns to a field after leaving it, and the generator never returns to an earlier question, since its jump rows are logged against the question just answered. Q-06 and S-10 equal one whenever every question receives time and every section is complete, which is always the case here. The evaluation is therefore silent on these eight features, neither positive nor negative. All ten are retained in the output for analysts but carry no weight in detection. The complete table for all forty features, with group means, standard deviations, rank-biserial values and adjusted p-values, is **Appendix C**.

6.3 Baseline comparison and component contribution

All three methods below are computed from the same raw files, the same generated features and the same ground truth labels inside a single script, so the comparison involves no manual step and can be re-run in one command.

Table 3. Full detector against two baselines representing current practice, n500 SurveyCTO benchmark, 100 anomalies in 500 interviews.

Method	Precision	Recall	F1
Full detector	1.0000	1.0000	1.0000
Duration-only IQR baseline	0.9565	0.6600	0.7811
V-10 rule (a run of 20 or more events without back-navigation)	0.4648	0.6600	0.5455

The duration-only baseline reaches acceptable precision but misses 34 of the 100 anomalies, every one of which has a total duration inside the Tukey fences. The V-10 baseline flags 142 interviews to find the same 66, giving 76 false positives, and falls well short of the precision criterion. Neither baseline meets the recall threshold of 0.75. This is the central result of the project: aggregate duration alone misses about a third of the injected anomalies that multi-feature analysis captures.

The same scrutiny is then turned inward. The combined detector flags an interview when its anomaly score reaches two, where the score is the number of active rule flags plus at most one point contributed jointly by the interquartile fences. That cap makes a threshold of two unreachable for the fences alone, so each component is isolated on its own terms: the interquartile component is credited when any fence fires, and the rule component is held to the same threshold of two that the combined detector applies, with the fence contribution removed.

Table 4. Contribution of each detection component, aggregated over the ten runs of the robustness round.

Component	Precision	Recall	F1	TP	FP	FN
Combined, as shipped	0.9920	0.9416	0.9662	871	7	54

Component	Precision	Recall	F1	TP	FP	FN
Rules only	0.9988	0.9319	0.9642	862	1	63
Interquartile fences only	0.6628	0.6162	0.6387	570	290	355

The rule layer carries the detection. On its own it reaches precision 0.9988 and recall 0.9319, within a point of the combined detector on both measures. The interquartile layer on its own reaches precision 0.6628 and recall 0.6162, and under the stress condition its recall falls to between 0.19 and 0.28. Combining the two remains worth doing, but the margin is small and should be stated as such: the combination recovers nine true positives the rules miss, lifting recall from 0.9319 to 0.9416, and costs six additional false positives, moving precision from 0.9988 to 0.9920. That is the entire measured contribution of the statistical component in the shipped configuration, and it is a smaller contribution than the emphasis the project proposal placed on interquartile thresholding would suggest.

The interquartile component also degrades in an instructive way. Across the prevalence sweep its recall holds at 1.0000 while anomalies make up 5, 10 and 20 percent of the sample, then falls to 0.6667 at 30 percent. This is masking: the fences are estimated from a sample that contains the outliers they are meant to isolate, so as contamination rises the quartiles move with it and the fences stop reaching the anomalies. Unsupervised methods of this kind assume that normal instances are far more frequent than anomalies, and Chandola, Banerjee and Kumar (2009, §3) warn that violating the assumption produces a high false alarm rate; here it runs the other way, costing recall rather than precision, because the contamination pulls the fence towards the anomalies instead of away from them. It is a property of applying unsupervised fences to contaminated data rather than a defect in the implementation, and the same mechanism accounts for part of the fabrication shortfall analysed in section 6.6.

The practical reading is that the rules, grounded directly in the behavioural signatures documented in the literature, do the heavy lifting, while the statistical component contributes marginal gains in conditions where class separation is already wide. Full per-run values, including confusion counts for each component, are regenerated by the command in Appendix D.4.

6.4 Class imbalance sweep

On canonical-difficulty datasets of five hundred interviews generated from a held-out seed, recall remained 1.0000 at every prevalence level while precision varied with prevalence as expected.

Table 5. Class imbalance sweep, held-out seed 7, canonical difficulty, SurveyCTO, 500 interviews.

Anomaly prevalence	Precision	Recall	F1
5 percent	0.9615	1.0000	0.9804
10 percent	1.0000	1.0000	1.0000
20 percent	1.0000	1.0000	1.0000
30 percent	1.0000	1.0000	1.0000

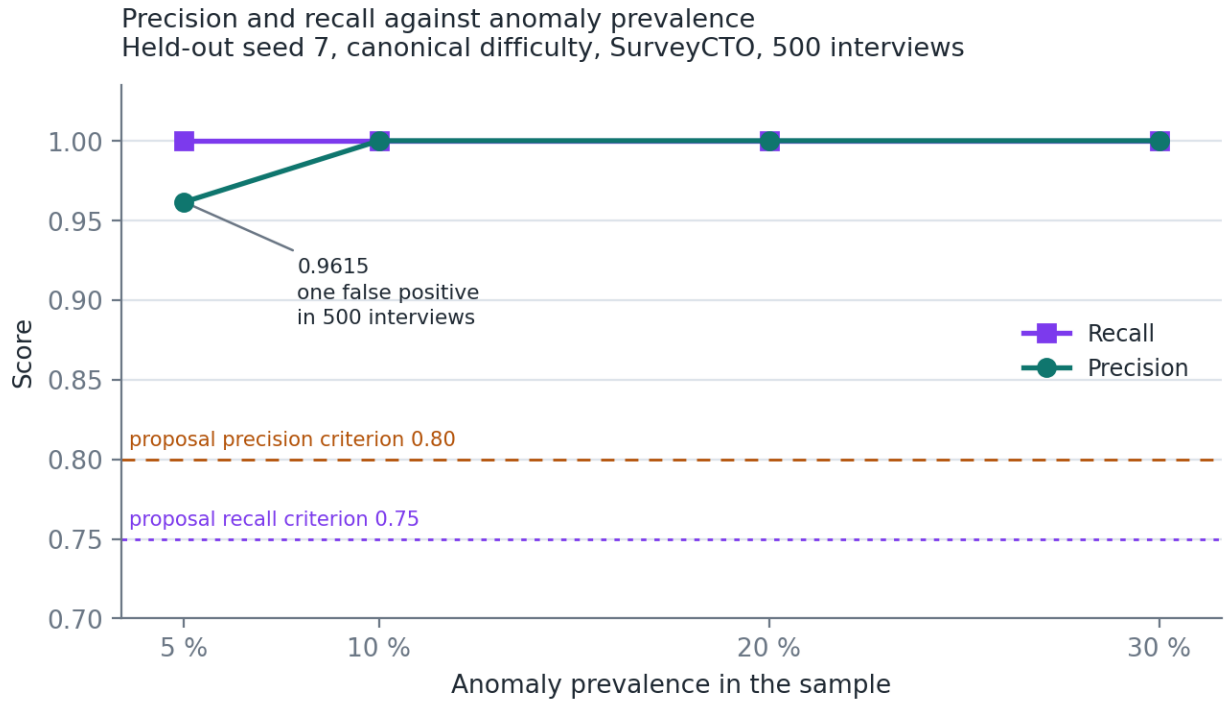


Figure 2. Precision and recall against anomaly prevalence. Recall holds at 1.0000 across the sweep. Precision is 1.0000 at 10, 20 and 30 percent prevalence and 0.9615 at 5 percent, where one false positive among twenty-five true anomalies costs almost four points. Both proposal criteria are cleared by a wide margin at every level.

At five percent prevalence a single genuine interview was falsely flagged, costing almost four points of precision. This sensitivity of precision to prevalence is a property of the metric rather than of the detector, and it matters operationally because real fabrication rates are usually low. On this sweep the detector raised one false alarm in 500 interviews at five percent prevalence. That is a single observation on generated data, not an estimate of field review burden, and section 7 treats it as a hypothesis for a field pilot to test.

6.5 Held-out stress evaluation

The stress condition is the demanding test. Profiles overlap by construction, seeds are held out, and detector settings are the defaults with no calibration to the evaluated data.

Table 6. Held-out stress evaluation, seeds 7 and 99 in both platform formats, 500 interviews each.

Dataset	Precision	Recall	F1
Seed 7, SurveyCTO	0.9770	0.8500	0.9091
Seed 7, ODK	0.9570	0.8900	0.9223
Seed 99, SurveyCTO	1.0000	0.8800	0.9362
Seed 99, ODK	1.0000	0.8400	0.9130

All four runs exceed both success criteria.

Both baselines were applied to the same four runs (Table 6a). The duration-only check flags no anomalous interview in any of them: under stress the anomalous profiles' total durations fall inside the Tukey fences, and the fences flag between zero and five genuine interviews instead. The V-10 rule recovers between 35 and 51 percent of anomalies at precision between 0.16 and 0.22, and over the four runs it finds 101 of 132 fabricated and 80 of 132 speeding interviews but none of the 136 disengaged ones, which navigate by design. This is the central result of section

6.3 under the harder condition: the gap between multi-feature analysis and aggregate duration widens, rather than narrows, when profiles overlap.

Table 6a. Both baselines on the held-out stress runs, beside the full detector. Source: `output/robustness_validation/baseline_breakdown.csv`, produced by `scripts/baseline_breakdown.py` from the recorded features of the robustness round.

Dataset	Full detector, P / R	Duration IQR, P / R	V-10 rule, P / R
Seed 7, SurveyCTO	0.9770 / 0.8500	0.0000 / 0.0000 (0 TP, 0 FP)	0.1613 / 0.3500
Seed 7, ODK	0.9570 / 0.8900	0.0000 / 0.0000 (0 TP, 5 FP)	0.2017 / 0.4800
Seed 99, SurveyCTO	1.0000 / 0.8800	0.0000 / 0.0000 (0 TP, 1 FP)	0.2117 / 0.4700
Seed 99, ODK	1.0000 / 0.8400	0.0000 / 0.0000 (0 TP, 1 FP)	0.2125 / 0.5100
All four runs	0.9830 / 0.8650	0.0000 / 0.0000	0.1974 / 0.4525

The first submission also read the similarity between the SurveyCTO and ODK rows of Table 6 as evidence that the two parsers are equivalent. That inference was not available. The generator was called once per format on an advancing random state, so the two files of each dataset held different interviews, and any difference between them mixed a parser effect with a sampling effect. To isolate the parsers, each dataset was regenerated with one history written into both formats (`scripts/run_platform_parity.py`). The SurveyCTO rendering is byte-identical to the file evaluated in the robustness round, so its results are the ones already reported.

Table 6b. One generated history written into both formats, 500 interviews per dataset. Source: `output/platform_parity/`.

Dataset	SurveyCTO, P / R	ODK, P / R	Interviews with any of the 40 features different	Interviews with a different score or category
Canonical, seed 42	1.0000 / 1.0000	1.0000 / 1.0000	0 of 500	0 of 500
Stress, seed 7	0.9770 / 0.8500	0.9770 / 0.8500	0 of 500	0 of 500
Stress, seed 99	1.0000 / 0.8800	1.0000 / 0.8800	0 of 500	0 of 500

On matched histories the two parsers produce identical features, scores and categories for all 1,500 interviews, and therefore identical confusion counts. This is the evidence for the common intermediate representation, and it is stronger than the original claim, because it shows equivalence interview by interview rather than similarity in aggregate. It also accounts for the differences between the SurveyCTO and ODK rows of Table 6 and Appendix B.2: they are differences between the sampled interviews, not between the parsers. The equivalence holds for the two synthetic formats; section 3.1 explains why it does not yet extend to native exports.

6.6 Detection by anomaly type

Per-class recall on the held-out stress data shows a clear gradient of difficulty. Each figure below is the recall for that injected class, that is the proportion of interviews of that type which the combined detector flagged, aggregated across the four stress runs. These are diagnostic breakdowns of a single detector rather than separate detectors, and the proposal's success criteria of precision 0.80 and recall 0.75 are defined for the combined detector, which exceeds both on every run. No committed target is missed by any figure in this section.

Table 7. Per-class recall by injected anomaly type, held-out stress runs. Each cell is the count of that class the detector flagged over the count present.

Run	Speeding	Disengaged	Fabricated

Run	Speeding	Disengaged	Fabricated
Seed 7, SurveyCTO	33/33	28/34	24/33
Seed 7, ODK	33/33	32/34	24/33
Seed 99, SurveyCTO	33/33	30/34	25/33
Seed 99, ODK	33/33	31/34	20/33
Aggregate	**132/132 (1.00)**	**121/136 (0.89)**	**93/132 (0.70)**

Speeding was detected in all 132 instances. Disengagement was detected in 121 of 136 instances, or 89 percent, driven by the median-relative pause rule combined with elevated back-navigation. Fabrication was detected in 93 of 132 instances, a per-class recall of 0.70, with per-run rates between 0.61 and 0.76.

The undetected fabricated interviews all came from the mildly compressed variant. The stress generator emits two fabricated profiles, and the tightly compressed one was detected without exception, so the aggregate figure combines one profile at perfect recall with another at approximately half. A bounded diagnostic pass on the false negatives, conducted on development seeds and reported in full in the accompanying validation documentation, established that these interviews are not in fact indistinguishable from genuine ones. Response time entropy separates the two populations completely on those datasets, the highest fabricated value being 1.983 against a lowest genuine value of 2.039. The misses arise from how the detector reads that feature rather than from an absence of signal in it: the behavioural entropy rule applies an absolute constant calibrated to the tightly compressed profile, and the corresponding interquartile fence is estimated from a sample that already contains twenty percent injected anomalies, which depresses the lower quartile and the fence with it. This second effect is masking in the sense Hubert and Vandervieren (2008, §1) use the term: the quantity that defines the fence is itself distorted by the anomalies the fence is meant to isolate.

A related structural limitation applies to the right-skewed non-negative features: for quantities such as response time variance the Tukey lower fence falls below zero and is therefore unreachable, so no low-side outlier can be detected on them at all, which matters because the fabrication signature is a low-side deviation on exactly those quantities. Section 8 develops this point. These are calibration and fence-construction limitations rather than evidence that

univariate analysis is insufficient for this profile, and they are recorded here as the honest account of where the method's recall is lost. The full per-run counts behind this section are in Appendix B.

A worked example. Interview INT-00009 in the seed 7 SurveyCTO stress run is one of the mildly compressed fabricated interviews, and shows how narrowly the detector reaches them. Its total duration of 80,785 ms is 0.48 of the sample median, just under the 0.5 cut-off, so the short relative duration flag fires. Its mean response time of 3,665 ms is below half the sample median of 7,576 ms, which is 3,788 ms, so the low mean response time flag fires. Its response time entropy of 1.236 bits is below the lower Tukey fence of 1.295 bits estimated from that run, so the entropy fence fires and contributes the one interquartile point. The score is one plus two, three, and the anomaly report lists the interview as suspicious with `flag_short_relative_duration, flag_low_mean_rt, flag_iqr_entropy` beside it. The absolute entropy rule does not fire, at 1.236 bits against 0.5, nor does uniform timing, at a coefficient of variation of 0.14 against 0.1, and the interview contains one jump, so the navigation rule is silent. Each of the three flags that did fire cleared its threshold by less than five percent, which is what a per-class recall of 0.70 looks like at the level of a single interview. The two false positives in the same run, INT-00287 and INT-00341, each scored exactly two, from moderate back-navigation and the variance fence, which is a flag pair a supervisor reading the listed flags could weigh quickly.

6.7 Scalability and regression checks

Feature extraction uses vectorised operations with time complexity approximately linear in the number of audit events, and the detection step is dominated by quantile computation over interviews. Scaling behaviour was measured rather than asserted, by running the same SurveyCTO pipeline over benchmarks of 500 and 10,000 interviews and recording both accuracy and wall clock runtime.

Table 8. Scale behaviour. Source: the strict examiner audit run, `output/examiner_audit/strict_examiner_audit_report.md` and its JSON companion, in which these times appear as `n500_runtime_seconds` and `n10000_runtime_seconds` under the scale stability check.

Dataset	Interviews	Runtime	Precision	Recall
`n500`, SurveyCTO	500	56.12 s	1.0000	1.0000
`n10000`, SurveyCTO	10,000	1,068.87 s	0.9945	1.0000

A twentyfold increase in input cost a factor of 19.05 in runtime on this run, and accuracy is essentially preserved: recall holds at 1.0000 and precision moves from 1.0000 to 0.9945, a loss of eleven interviews to false positives in ten thousand. The runtime factor itself should not carry weight, because it moves between runs and machines: 18.04 in a second run on the same machine, described below, 19.97 in the clean-environment rerun recorded in Appendix D, and between 20.3 and 21.0 in the independent reproduction on other hardware. What holds across all of them is that the cost per interview stays roughly constant, about 0.11 s at both sizes on this machine, which is the near-linear scaling the design targets.

Memory was measured rather than assumed, with each dataset processed in a fresh process by `scripts/measure_memory.py` (Table 8a). The first submission stated that only interview-level summaries are held for the batch stage. That is not what the code does: the batch path used by `run_pipeline.py` keeps the question-level and section-level tables for every interview in memory until they are written out, alongside the interview summaries, so memory grows linearly with the number of interviews. Above an interpreter and library baseline of about 142 MiB, the pipeline added about 19 MiB for 500 interviews and 68 MiB for 2,000, roughly 34 KiB per interview, and 337 MiB for 10,000, which is the same rate. Memory is not a constraint at these sizes, but a survey of several hundred thousand interviews would need the granular tables streamed to disk rather than held.

Table 8a. Peak resident memory by dataset. Sources: `output/memory_profile.md` and, for n10000, `output/memory_n10000/memory_profile.md`; Windows 11, Python 3.14.0 with pandas 2.3.2. In the pinned environment of Appendix D.1 (pandas 3.0.1) the same measurement

gave 172.1 and 173.9 MiB at n500 and 247.7 and 247.5 MiB at n2000; the rate per interview is higher there, about 50 KiB, but growth is still linear.

Dataset	Interviews	Question-level rows	Peak memory	Above baseline
`n500`, SurveyCTO	500	10,000	161.1 MiB	19.0 MiB
`n500`, ODK	500	10,000	160.1 MiB	18.5 MiB
`n2000`, SurveyCTO	2,000	40,000	210.0 MiB	68.1 MiB
`n2000`, ODK	2,000	40,000	209.5 MiB	67.4 MiB
`n10000`, SurveyCTO	10,000	200,000	479.1 MiB	337.4 MiB

The runtime figures above are wall clock timings of a single run and they move between runs on the same machine. A separate aggregate timing file, `output/synthetic_dataset_metrics.csv`, records 57.54 s and 1,037.74 s for the same two datasets, a factor of 18.04, from a different run. These are two measurements, not one measurement reported twice, and the strict examiner audit is the source cited throughout this report and in the validation documentation. Both runs, and the independent reproduction, show a roughly constant cost per interview, which is the claim the scaling result supports; whether the factor lands just below or just above twenty is variation between runs.

After all changes made during the robustness round, the canonical benchmark suite continued to score precision 1.0000 and recall 1.0000 in both platform formats, confirming that hardening the evaluation did not degrade performance on the original success cases, and the automated test suite, now forty-three tests, passed in full.

7. Discussion

The results support three substantive conclusions. First, the common intermediate representation works as designed for the formats it was tested on: when one generated history is written in both synthetic formats, the two parsers produce identical features and identical decisions for every

interview (Table 6b). Whether that extends to the platforms' native exports is untested (section 3.1). Second, multi-feature timing analysis clearly outperforms the aggregate-duration checking that represents current practice, raising recall from 0.6600 on the canonical benchmark, and from zero under stress, to between 0.84 and 1.00 while holding precision at or near 1.00. The features doing the heaviest work for fabrication are the distributional shape measures, which is what the falsification literature reports (section 2.3). The pause and navigation measures that carry disengagement have no such backing: as section 5.1 records, that profile is a design assumption rather than a documented signature. In both cases, as section 5.2 notes, these are also the measures the generator was written to move.

The closest comparable work is Birnbaum et al. (2013), and the relationship between their results and these should be stated carefully. They instrumented ODK Collect, the platform this project parses, and trained classifiers on behavioural features to identify fabricated interviews, reaching up to 96 percent accuracy, and 86 percent against interviewers who knew the method was in use and were rewarded for evading it (section 2.3). Three differences make the detection levels incomparable. Their labels come from human behaviour, obtained by having interviewers fabricate under controlled conditions, whereas the labels here come from a generator whose profiles encode this project's own assumptions. Their method is supervised, so it learns which combinations of features separate the classes in their data, whereas the detector here applies rules that were fixed before the evaluation data existed. And their unit of comparison is the individual interviewer, since their strongest features are normalised against that interviewer's own mean, whereas this project normalises against the sample median, because an audit log does not always carry an enumerator identifier.

What can be compared is which signals carry the result, and what it costs to work without labels. Their finding that the behavioural data is what does the work, with accuracy falling to little better than chance without it, matches the component breakdown in section 6.3, where the rules reading behavioural features carry the detection and the statistical fences add at the margin. Their interviewer-normalised features are the counterpart of the median-relative thresholds in Table 1a, and that two independent designs converged on relative rather than absolute comparison is the clearest point of agreement between them. The contribution here is not a better detection rate, which these two designs cannot establish against each other, but an operational one: the same

family of signals, computed automatically from either of two platform formats through a documented and tested pipeline, with the thresholds, the baseline comparison and the failure modes all reported, and without the labelled training set that a survey organisation is unlikely to possess.

Third, and equally important, the evaluation shows where the approach stops working. Early runs of this project scored perfect precision and recall everywhere, which was initially read as success. Closer inspection showed the synthetic anomalies were too cleanly separated and parts of the detector configuration mirrored the generator. Treating perfect scores as a warning rather than a result, and rebuilding the evaluation around held-out seeds, overlapping profiles and prevalence sweeps, produced lower numbers and more defensible results.

The residual misses concentrate in mild fabrication. My initial reading was that these interviews sit inside normal bounds on every individual feature and are therefore beyond the reach of any univariate method. The diagnostic pass reported in section 6.6 does not support that reading. Response time entropy separates the missed fabrications from genuine interviews with a clear margin and no overlap, and the detector fails to act on that separation because the rule reading entropy uses a fixed constant fitted to a different fabrication profile, and because the accompanying fence is estimated from data that already contains the anomalies it is meant to isolate. The shortfall is therefore a calibration and estimation problem within the univariate design, not proof that the design has reached its ceiling.

That distinction matters for what should be done next: a corrected threshold is a smaller and better understood change than a move to multivariate modelling, and it was therefore attempted first. The correction retains the absolute constant as a floor and adds a fence at the median minus three times the distance from the median to the third quartile, $M - 3(Q3 - M)$. The scale is taken from the upper half of the entropy distribution because every anomaly profile in scope lowers entropy and none raises it, so that half is less contaminated than the lower half. It is not uncontaminated, as the first submission said: the anomalies sit below the median and pull it down, and if every anomaly lies below every genuine interview, twenty percent contamination moves the sample median to the 37.5th percentile of the genuine interviews. Nor does the fence reduce to Tukey's, as the first submission also said. For a symmetric distribution with $Q3 - M = d$, the interquartile range is $2d$ and Tukey's lower fence is $Q1 - 1.5 \times 2d = M - 4d$, whereas the

correction gives M − 3d. It is tighter by d, and equivalent under symmetry to a Tukey fence with a multiplier of 1.0 rather than 1.5, so some of the additional detection and some of the additional false positives reported below follow from the tighter fence rather than from the change of scale estimate alone. It was developed on the development seeds, frozen, and evaluated once against the held-out data. Detection of fabricated interviews rose from 93 of 132 at 0.70 to 132 of 132, and the lowest precision across the round fell from 0.9570 to 0.8333. The gain concentrates in the stress condition and the cost concentrates at low prevalence, where recall was already complete and the additional flags fall entirely on genuine interviews. Since realistic fabrication rates are low, the change as it stands trades false accusations for no additional detection in the condition that most resembles deployment. It is therefore held on a separate branch and is not part of the evaluated system reported here. Whether to adopt it is a judgement about the relative cost of an undetected fabrication against a genuine enumerator sent for review, which is a deployment decision rather than a methodological one.

The severity categories, suspicious at a score of two and anomalous at five, translate the score into workflow terms: suspicious interviews warrant a supervisor review while anomalous interviews warrant investigation before the data is used. Because the score is a count of named flags, a supervisor can see exactly why an interview was flagged, a property that separates this design from opaque model-based alternatives and that matters wherever a flag has to be explained to the person whose work it concerns (Goel et al., 2022, p. 11).

A practical deployment would run the pipeline against each day's uploaded audit logs, review the anomaly report each morning, and route suspicious interviews to back-checking while quarantining anomalous ones. How much review effort that creates is a hypothesis, not a result. The one false alarm in 500 interviews at five percent prevalence comes from a single synthetic run, and a field pilot with back-checked outcomes would be needed before it could inform staffing.

Because a flag concerns a named enumerator's work, a deployment would also need a documented process for adjudicating and challenging flags. A flag should open a review, never conclude one. The supervisor sees the listed flags and their values against the thresholds, as in the worked example in section 6.6, and checks them against independent evidence such as a back-check visit or, where available, GPS or audio audit. The enumerator should be told which

signals fired and given the chance to explain them, since long pauses, fast but valid respondents and an absence of navigation all have innocent causes. The outcome should be recorded, so that rules which repeatedly fire on interviews later found to be genuine can be recalibrated, and no decision about an enumerator's employment should rest on the score alone. The per-flag reporting also supports feedback to enumerators, since a pattern of speeding flags concentrated in one team is actionable in a way that a bare quality score is not.

8. Limitations

Evaluation is synthetic throughout. The generator was built to mimic documented properties of real audit logs, including structural events, complexity-correlated timing and realistic navigation, but no synthetic process reproduces the full variability of field data, and the results here are best read as an upper bound on operational performance, because the injected anomalies match the detector's assumptions by construction (section 5.2). Two consequences deserve naming rather than leaving implicit. Eight of the forty features are constant across the synthetic corpus, as reported in section 6.2, so this evaluation says nothing about their discriminative value in either direction; it is silent on them, not negative. And the anomaly profiles are the ones the generator was written to inject, which means the evaluation measures the detector against a documented threat model rather than against the full space of behaviours a real enumerator might produce, and because the same author wrote the generator and the detector, the two share one set of assumptions about what each anomaly looks like. Validation against real interviews with independently verified outcomes remains the most important extension.

Detection is univariate. Anomalies expressed only in joint feature patterns will evade fence-based and rule-based flags. The fabrication shortfall reported in section 6.6 is not an instance of that limit, since the diagnostic found a single feature that separates the missed cases cleanly and with no overlap, but the limit remains real for an adversary whose individual features all fall within normal ranges while their combination does not.

A structural asymmetry in the Tukey fences. For strongly right-skewed non-negative quantities such as response time variance and interquartile range, the lower fence at the first quartile minus one and a half interquartile ranges falls below zero and is therefore unreachable, so no low-side outlier can ever be flagged on those features. This matters because the fabrication

signature is a compression, that is a low-side deviation, on precisely those quantities. Response time entropy escapes the problem because it is bounded and roughly symmetric, which is why it is the feature that separates the missed fabrications. The consequence is that the statistical layer is structurally weaker against compression than its formulation suggests, and this should be read alongside the component breakdown in section 6.3 rather than separately from it.

No platform asymmetry once histories are matched. The first submission reported that the ODK path returned lower precision and recall than the SurveyCTO path across the robustness round, stated that all four false positives in the round arose on ODK inputs, and attributed the difference to the ISO 8601 conversion step. Two parts of that were wrong. The round raised seven false positives, not four: four on ODK inputs, all in the seed 7 stress run, and three on SurveyCTO inputs, two in the same stress run and one in the five percent sweep (Tables 4 and B2). And the comparison was not made on matched conditions. The ODK and SurveyCTO files of each dataset held different generated interviews, and the prevalence sweep, which supplies 2,000 of the 3,500 SurveyCTO interviews in Table B2, was run in SurveyCTO format only. When the same history is written into both formats the difference disappears entirely (Table 6b), so the aggregate asymmetry in Table B2 reflects which interviews each format happened to receive, and no parser mechanism is needed to explain it. Platform comparisons in this report are now made only on matched histories.

No native export has been processed. As section 3.1 records, the formats the evaluation ran on are the project's synthetic layouts, not the formats the platforms export natively. ODK support has since been extended to the documented native layout, with detection and form-level event handling covered by a test, but that test is built from the documentation rather than from a real export, and SurveyCTO's native text audit remains unsupported in both its forms. Every result in section 6 was therefore produced on synthetic data in synthetic formats. Until a real export from each platform has been parsed and checked, the claim of platform support rests on the documentation and on two synthetic layouts, not on the platforms themselves.

Threshold provenance and instrument dependence. The thresholds shipped as defaults were chosen from the literature and from calibration data. Every threshold is configurable (Table 1a), and six of the fourteen rules are expressed relative to the sample median, but deployments on

instruments very unlike the twenty-question household pattern of the benchmarks should recalibrate rather than assume the defaults transfer.

External dependencies. The pipeline depends on the integrity of device clocks and on each platform continuing to emit its current audit schema. Clock drift or schema changes would degrade timestamps or parsing respectively; the parsers detect and mark such records rather than repair them, so the failure is visible in the quality flags rather than silent in the features.

9. Conclusion

This project set out to make timing-based paradata usable for survey quality assurance at scale, and delivered a working system: format-aware parsers for the two dominant collection platforms, a documented forty-feature library spanning three levels of granularity and a statistical characterisation layer, a transparent and configurable anomaly detector, a labelled synthetic benchmark generator, and an evaluation that meets its predefined success criteria under deliberately hostile conditions. Precision remained between 0.96 and 1.00 and recall between 0.84 and 1.00 on data the detector had never seen, including a stress condition constructed to blur the line between genuine and anomalous behaviour. The system detects speeding essentially perfectly, disengagement reliably, and blatant fabrication reliably. Mild fabrication is detected in roughly seven cases in ten, and the diagnostic reported here locates that shortfall in threshold calibration rather than in the univariate design itself, which makes it a tractable target rather than a boundary. Five of the six objectives set at proposal stage are met in full. The second, format-aware parsers for SurveyCTO and ODK, is met for the two synthetic formats evaluated here and only in part for the platforms' native exports (section 3.1). Future work should validate the parsers on native exports and the system against real field data with verified outcomes, settle the entropy fence trade-off against operational evidence of relative error costs rather than synthetic prevalence alone, extend detection with an interpretable multivariate layer, and add parsers for further platforms, each of which the architecture was explicitly designed to accommodate.

9.1 Reflection

Three things from this project will stay with me.

The first is what the early results taught me about evaluating my own work. The first full evaluation returned perfect precision and recall on every dataset, and I recorded it as success. It was not: the synthetic anomalies were too cleanly separated from genuine interviews, and parts of the detector had been tuned to the generator that produced them. Rebuilding the evaluation around held-out seeds, overlapping profiles and a prevalence sweep produced lower numbers, and those lower numbers are the ones I can defend. I have stopped reading a perfect score as a result and started reading it as a warning that the test is too easy.

The second is the entropy fence. The diagnostic showed that the fabricated interviews the detector misses are separable on response time entropy, and the correction I developed raised fabricated recall from 93 of 132 to 132 of 132. It also dropped the lowest precision in the round from 0.9570 to 0.8333, and those additional false positives fall on genuine interviews at the low prevalence that most resembles a real survey. Every flag in this system is an accusation about a named enumerator's work, so I did not merge it. Deciding against my own improvement, and reporting both numbers, was the hardest judgement in the project, and I still think it was the right one.

The third is about sources, and it is not comfortable. Two of the references in my first submission could not be located, and they carried some of the claims I leaned on most heavily. Losing them cost me the fabrication signature, the disengagement signature and seventeen of the forty feature justifications, and it taught me that a citation is a claim about a document that either exists and says what I attribute to it, or does not. I now check every reference against its primary source, record the check with a date, and treat a source I have not read myself as unusable. With the time again I would keep that record from the first week rather than reconstructing it at the end, and I would log every use of an AI tool as it happened, for the same reason: a record made afterwards is weaker evidence than one made at the time.

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Appendix A. Artificial intelligence usage disclosure

A.1 Purpose and scope

This appendix is the full disclosure of artificial intelligence tool use in this project, required by section 9.4 of the project proposal and by the supervision framework for the academic year. It is maintained in the repository as `docs/ai_usage_log.md` and reproduced here in expanded form. The log was started on 6 July 2026; entries before that date are reconstructed from the development history and version control record and are marked as reconstructed, which is a limitation discussed in section A.7.

The intellectual content of this project is my own. The research question, the literature review and the argument it makes, the forty-feature taxonomy, the common intermediate representation, the parser design, the choice of interquartile and rule-based detection, the thresholds shipped on `main`, the synthetic generator and its anomaly profiles, the success criteria, the interpretation of every result, and the decision to report weaker per-class results transparently are mine.

A.2 What was provided to external tools, and what was not

No respondent data, personal data, or third-party data of any kind was uploaded to any external tool at any point in this project. This is not a matter of policy alone: all evaluation data in this project is synthetic, generated by `tests/generate_benchmark_datasets.py` from recorded random seeds, and describes interviews that never took place with people who do not exist. There is no real survey data anywhere in the repository or in the evaluation.

The categories of material provided to tools were the following, and no others:

Category	Examples	Sensitivity
Project source code	Parser classes, feature extractors, detector, test files	Mine; public in the repository
Project documentation	Draft report and validation text, progress reports	Mine; public in the repository
Synthetic output files	Metrics CSVs, confusion counts, feature datasets	Machine generated; no human subject

Category	Examples	Sensitivity
Error messages and stack traces	pandas and pytest output	None
Published literature summaries	My own paraphrases of cited works	None

A.3 Tools and versions

Version identifiers are recorded where they are recoverable. Hosted assistant products do not expose a stable version string to the user and were not pinned at the time of use, so those entries state the product and the period of use rather than a version number, which is the most precise honest statement available.

Tool	Version	Recoverable	Period of use
GitHub Copilot	Visual Studio Code extension, version not recorded at time of use	No	March to April 2026
Claude (hosted assistant)	Product version not exposed to the user; not pinned	No	March to September 2026
ChatGPT (hosted assistant)	Product version not exposed to the user; not pinned	No	April 2026
Python	3.14.0	Yes	Final validation runs
pandas	3.0.1	Yes	Final validation runs
Full environment	`requirements-validated.txt`, 143 pinned packages	Yes	Final validation runs

The last three rows are not AI tools and are listed because the reproducibility of every figure in this report depends on them, and because two of the five defects recorded in section 4 were caused by a pandas version change rather than by a logic error.

A.4 Log of use, by episode

Each row records what the tool was asked for in substance, what was given to it, and what happened to the output in the submitted work.

Table A1. Artificial intelligence tool use, by episode.

Date	Tool	Representative prompt, in substance	Input provided	Outcome in submitted work
Mar 2026 (reconstructed)	GitHub Copilot	Inline completion while writing parser and DataFrame code	Open source files in the editor	**Modified.** Line-level completions only. Logic was written, reviewed and tested by me.
Mar 2026 (reconstructed)	Claude	Clarify the difference between Tukey interquartile fences and the modified z-score for right-skewed data, and which is defensible for response times	Prose question, no code, no data	**Accepted as background.** The method choice is mine and is unchanged. The statement in the first submission that the citations supporting it had been checked against their sources was wrong: two of the references carrying that argument could not be located when they were checked in September 2026, and both have been withdrawn. Appendix E records the checks.
Mar to Apr 2026 (reconstructed)	Claude	Why does timestamp parsing produce different magnitudes for epoch and ISO inputs, and what guard would catch it	Parser code and an error trace	**Modified.** The suggestion of a magnitude-based guard for epoch seconds was useful; I implemented, named and tested the behaviour myself.

Date	Tool	Representative prompt, in substance	Input provided	Outcome in submitted work
Apr 2026 (reconstructed)	ChatGPT	Review the structure of a labelled evaluation for label leakage	Description of the evaluation join, no data	**Accepted as a check.** Confirmed the detector must not see labels. The evaluation code and the strict one-to-one join were written by me.
6 Jul 2026	Claude	Review `src/` for defects and comment on whether the generator is realistic enough for the claims made	Source tree and generator code	**Accepted in part.** I fixed the confirmed defects, removed detector settings that mirrored the generator, and re-ran on held-out seeds. Comments on realism I judged speculative were rejected.
6 to 14 Jul 2026	Claude	Drafting support for progress documentation; run clean-environment checks and report what breaks	Documentation drafts, repository	**Modified.** All draft text was rewritten by me. The clean-run checks surfaced two real defects, described in section 4, which I fixed and covered with tests.
19 Aug 2026	Claude	Check every figure quoted in the documentation against the file it claims to come from, and check that tags and links	Documentation and `output/` files	**Accepted after verification.** Corrections were applied only where I confirmed the discrepancy against the recorded output myself.

Date	Tool	Representative prompt, in substance	Input provided	Outcome in submitted work
		resolve		
19 to 25 Aug 2026	Claude	Identify what the validation documentation is missing relative to the proposal commitments	Validation documentation, `output/`	**Accepted in part.** This identified the absence of aggregate confusion counts, baseline figures, effect sizes and a per-component breakdown. Every value added was recomputed from existing project outputs by scripts in `scripts/` and checked by me.
19 to 25 Aug 2026	Claude	Bounded diagnostic on the fabricated false negatives: is there a feature that separates them	Development seeds 11 and 42 only, synthetic	**Accepted as diagnosis, rejected as a change.** The finding that entropy separates the missed cases completely is reported in section 6.6. The threshold correction it motivated was developed, evaluated once, and deliberately not merged; it remains on a separate branch. No threshold on `main` was changed.
25 Aug to 5 Sep 2026	Claude	Locate stale or contradictory wording across the report, validation document and presentation; assemble front matter, figures and appendices	Documentation and `output/` files	**Modified.** Corrections were made against recorded outputs. The figures in sections 4, 6.4 and 6.6 are generated by `scripts/build_report_figures.py` from values transcribed from recorded output files, and I checked each plotted value against its source.
10 to 16 Sep 2026	Claude Code (Claude Opus 5)	Check every reference against its primary source, obtain and read	The repository, the report, and public sources fetched from the web	**Modified.** The tool carried out the bibliographic checks, read the sources that were obtainable, and drafted the rebuilt literature review, the comparison in section 7, appendices E and F and the

Date	Tool	Representative prompt, in substance	Input provided	Outcome in submitted work
		the open-access full texts, and rebuild section 2 from them; draft the corrections the supervisor's review required		reflection in section 9.1. Appendix E records how far each source was verified; no claim in section 2 rests on a source listed there as abstract-level or not consulted. The drafted prose is mine to read, correct and endorse before submission, and section A.5 states what that leaves outstanding.
10 to 16 Sep 2026	Claude Code (Claude Opus 5)	Make the fence multiplier configurable, add the regression tests the review found missing, and build the baseline, platform parity and memory analyses it asked for	The repository	**Accepted after verification.** Every change is covered by a test or reproduces a recorded output: the suite is now forty-two tests, the baseline script reproduces Table 3 exactly, the platform parity check is anchored to a SurveyCTO rendering that is byte-identical to the file the robustness round evaluated, and the whole of Appendix D was run in a clean environment with every recorded number reproducing.

A.5 The decisions that mattered

Three episodes changed the submitted work in ways worth stating explicitly rather than leaving inside a table.

Accepted. The clean-environment checks in July 2026 surfaced two defects that would otherwise have reached submission: the ISO 8601 to epoch conversion assumed the nanosecond datetime resolution of earlier pandas versions and silently produced values a thousand times too small under pandas 3, and the question-level response time was being computed as a first-to-last

event span rather than as the documented sum of event durations. Both are described in section 4. I accepted the finding, wrote the fixes and the tests, and both are covered by the suite.

Modified. The review of `src/` in July 2026 identified a coupling between two detector flags that matched the behaviour of an earlier generator. I accepted that the coupling was a form of overfitting and removed it, but I generalised the affected rules myself rather than adopting the specific replacement suggested, and I re-evaluated on held-out seeds rather than on the data the coupling had been fitted to.

Rejected. The entropy threshold correction motivated by the August diagnostic raises fabricated per-class recall from 93 of 132 at 0.70 to 132 of 132 and drops the lowest precision anywhere in the round from 0.9570 to 0.8333. I developed it, froze it, evaluated it once, and decided not to merge it, for the reasons argued in section 7: the cost concentrates at low prevalence, which is the condition that most resembles deployment. It is unmerged on a separate branch. Every figure in this report describes the detector without it.

Disclosed. The revision of September 2026 used AI assistance more extensively than any earlier episode, and the extent is stated here rather than left to the table. The tool checked every reference against its primary source, obtained and read the sources that were publicly available, rebuilt section 2 from them, drafted the comparison in section 7, appendices E and F and the reflection in section 9.1, wrote the three analysis scripts and the regression tests added in section 4, and applied the corrections the supervisor's review listed. Two boundaries were held throughout. No claim was written about a source that had not been read: where a source could not be obtained, Appendix E says so and the claim is held at the level the abstract or publisher record supports. And no number was taken on trust: every figure in the new tables is reproduced from a file under `output/`, and the whole of Appendix D was re-run in a clean environment to confirm it. What remains mine, and what the next paragraph does not excuse, is the obligation to read the drafted prose, confirm that it says what I am prepared to defend, and obtain the six sources still listed in Appendix E as unread.

A note on the entropy branch. The branch holding the entropy fence correction, `fix/entropy-threshold-relative`, was created from a rewritten history and shares no commit with `main`, so it cannot be merged by fast forward and is preserved as a separate line of development. Its code

comment claimed that the correction reduces to Tukey's fence for a symmetric distribution; that is wrong, for the reason given in section 7, and the comment has been corrected. The correction is a lower-side analogue of the semi-interquartile fence documented by Hubert and Vandervieren (2008, §1), which is discussed in section 2.5.

A.6 Tests attached to significant outputs

No code-level suggestion entered the submitted work without a test or a recorded output check behind it. The five defects described in section 4 are each covered by a test in the suite of forty-three, which passes in full on Python 3.14.0. The first submission made that claim of a suite of thirty-seven, and it was wrong: two of the five fixes, the ODK placeholder and the exclusion of administrative events from section statistics, had no test at all, and a third covered the preserved integrity flag only indirectly. The three tests added in September, named in section 4, make the statement true. Every quantitative claim in this report resolves to a file under `output/`, and the commands that regenerate those files are in Appendix D. The feature coverage table is generated by `scripts/build_feature_coverage.py` rather than maintained by hand, so it cannot drift from the code. The component breakdown in section 6.3 is recomputed from recorded predictions rather than from a fresh run, so it describes exactly the runs reported elsewhere in this report.

A.7 Limitations of this disclosure

Three limitations should be stated. First, entries before 6 July 2026 are reconstructed from the development history rather than recorded at the time, so their dates are approximate and their prompt summaries are reconstructions of substance rather than transcripts. Second, hosted assistant products do not expose a stable version identifier, so the tool versions for those entries are not recoverable and the log says so rather than inventing a number. Third, a log of this kind records episodes of use rather than every keystroke; inline editor completions in particular are continuous and were not individually recorded, and section A.4 characterises them as a class rather than enumerating them. Where the record is uncertain I have preferred to say so.

Appendix B. Per-run confusion counts

These are the raw counts behind every precision and recall figure in section 6. They cover the ten runs of the robustness round: two canonical benchmarks, four held-out stress runs, and four prevalence sweep runs.

B.1 Per run

Table B1. Confusion counts per run, robustness round.

Run	Format	TP	FN	FP	TN	Precision	Recall	F1
benchmark_n500_odk	ODK	100	0	0	400	1.0000	1.0000	1.0000
benchmark_n500_surveycto	SurveyCTO	100	0	0	400	1.0000	1.0000	1.0000
stress_seed7_odk	ODK	89	11	4	396	0.9570	0.8900	0.9223
stress_seed7_surveycto	SurveyCTO	85	15	2	398	0.9770	0.8500	0.9091
stress_seed99_odk	ODK	84	16	0	400	1.0000	0.8400	0.9130
stress_seed99_surveycto	SurveyCTO	88	12	0	400	1.0000	0.8800	0.9362
sweep_r005	SurveyCTO	25	0	1	474	0.9615	1.0000	0.9804
sweep_r010	SurveyCTO	50	0	0	450	1.0000	1.0000	1.0000
sweep_r020	SurveyCTO	100	0	0	400	1.0000	1.0000	1.0000
sweep_r030	SurveyCTO	150	0	0	350	1.0000	1.0000	1.0000

The lowest single precision anywhere in the round is 0.9570, on seed 7 in ODK format at stress difficulty. The lowest single recall is 0.8400, on seed 99 in ODK format. Both clear the proposal criteria of 0.80 and 0.75.

B.2 Aggregated by platform format

Table B2. Confusion counts aggregated by platform format.

Format	TP	FN	FP	TN	Precision	Recall
ODK	273	27	4	1,196	0.9856	0.9100
SurveyCTO	598	27	3	2,872	0.9950	0.9568

Both formats clear both criteria. The two rows are not a like-for-like comparison: the formats' files hold different generated interviews, and the prevalence sweep was run in SurveyCTO format only. On matched histories the two formats give identical results (Table 6b, section 8).

Source for both tables: `output/robustness_validation/`, summarised in `output/robustness_validation/robustness_summary.md`.

Appendix C. Full feature validity results

All forty catalogued features, tested with Mann-Whitney U on the n500 SurveyCTO benchmark of 400 normal and 100 anomalous interviews, with p-values Bonferroni-adjusted at a family-wise alpha of 0.05. Ordered by absolute effect size. Thirty features are significant after adjustment.

Seven features are constant across the entire corpus. For them the output file records a Mann-Whitney statistic of 20000.0, which is exactly 400 × 100 / 2, the value produced when every normal-anomalous pair is tied. The p-value is undefined because the statistic then has no variance; in this environment (SciPy 1.17.1) it is written as missing, and the independent reproduction recorded it as 1.0. The table shows it as not defined. S-04 is not quite constant, because its per-section ratios average to one only up to floating-point rounding, which gives it a statistic of 19998.5 and p = 1.0. Section 6.2 explains each of these features by its implementation; the evaluation is silent on their value in real data.

Table C1. Mann-Whitney U results for all forty features.

Feature	Cohen's d	Rank-biserial	Adjusted p	Significant
`T-03_rt_entropy`	-10.5664	1.0000	1.6e-52	yes
`Q-10_first_to_last_interaction_duration`	-4.1410	0.9983	3e-52	yes
`T-06_rt_mean`	-4.1370	0.9983	3e-52	yes
`S-01_mean_section_pace`	-4.1370	0.9983	3e-52	yes

Feature	Cohen's d	Rank-biserial	Adjusted p	Significant
Q-01_response_time_ms	-4.1370	0.9983	3e-52	yes
Q-02_time_to_first_interaction	-4.0951	0.9983	3e-52	yes
S-02_section_duration_ms	-3.8040	0.9942	8.2e-52	yes
V-09_session_length_vs_median	-3.7853	0.9930	1.1e-51	yes
V-01_total_interview_duration_ms	-3.7853	0.9930	1.1e-51	yes
T-02_rt_iqr	-3.5061	1.0000	2e-52	yes
T-07_rt_median	-3.3502	0.9625	1.4e-48	yes
T-08_rt_cv	-3.3206	0.9821	1.5e-50	yes
T-10_rt_max	-3.1625	1.0000	2e-52	yes
V-03_speed_trajectory_late	-2.7656	0.8933	7.3e-42	yes
V-02_speed_trajectory_early	-2.1527	0.6485	4.2e-22	yes
T-01_rt_variance	-1.8733	1.0000	2e-52	yes
S-03_within_section_variance	-1.7179	1.0000	2e-52	yes
T-04_rt_skewness	-1.4983	0.8032	7.2e-34	yes
V-10_zero_back_nav_flag	1.1444	0.4700	5e-19	yes
V-04_speed_trajectory_ratio	-1.0854	0.6551	1.5e-22	yes
Q-07_idle_time_ms	0.8919	0.2705	0.0011	yes
Q-09_constraint_trigger_count	-0.8874	0.4600	2.9e-15	yes
T-05_rt_kurtosis	-0.7455	0.5142	7.1e-14	yes
S-07_mean_pause_duration_ms	-0.4893	0.3200	2.9e-05	yes
V-05_back_navigation_rate	0.4048	0.2296	0.013	yes
V-06_back_navigation_total_count	0.3778	0.2298	0.011	yes
S-08_max_pause_duration_ms	-0.2172	0.3200	2.9e-05	yes
T-09_rt_min	-0.1918	0.3107	6.1e-05	yes

Feature	Cohen's d	Rank-biserial	Adjusted p	Significant
Q-03_edit_count	0.0838	0.2747	0.00065	yes
Q-08_keystroke_count	0.0838	0.2747	0.00065	yes
V-07_time_of_day_hour	0.0630	0.0347	1.0	no
V-08_day_of_week	-0.0596	0.0316	1.0	no
S-04_section_to_section_pace_ratio	0.0000	0.0001	1.0	no
S-05_pause_count_5s	0.0000	0.0000	not defined	no
S-06_pause_count_30s	0.0000	0.0000	not defined	no
S-09_back_nav_count_section	0.0000	0.0000	not defined	no
S-10_section_completion_rate	0.0000	0.0000	not defined	no
Q-04_revisit_flag	0.0000	0.0000	not defined	no
Q-05_revisit_count	0.0000	0.0000	not defined	no
Q-06_field_completion_ratio	0.0000	0.0000	not defined	no

Source: output/feature_validation_results.csv and output/feature_validation_results.txt, which additionally record group means, standard deviations and the raw test statistics. Regenerated by the command in Appendix D.5.

Appendix D. Reproduction commands

Every command below is run from the repository root. The state reported here is the tag `v-resubmission`, which contains every script this appendix names; `git rev-parse` on that tag resolves it to the exact commit. The earlier tag `v-validation-complete` predates `requirements-validated.txt` and `scripts/component_breakdown.py` and does not contain everything named here. Appendix D was run top to bottom on a copy of the working tree in a fresh virtual environment on 10 September 2026 (Windows 11, Python 3.14.0, `requirements-validated.txt`). Every step passed except the full workflow, which failed because the n10000 pack did not yet exist; the order above corrects that. Every recorded confusion count reproduced exactly, as did the component, baseline and parity tables, and the feature validity values agreed to within 6×10^{-16}. D.11 was not run there, because it requires Microsoft Word. The sequence was then run again on 16 September 2026 from a clean clone of the tagged tree, at commit `a432441`, which differs from the tag only in the rebuilt Word and PDF deliverables and in the page numbers measured into the contents lists. All fourteen steps passed, in one hour and fifty-three minutes, and every recorded number reproduced with no mismatches: the confusion counts of all ten runs of the robustness round, the component breakdown, the baseline table, the parity metrics and summary, and precision and recall of 1.0000 on the canonical suite. The feature validity values agreed to within 6 x 10^-16, the residual difference being the pandas version. D.11 was run on the submission machine and produced the Word and PDF deliverables that accompany this report.

D.1 Environment

```
python -m pip install -r requirements-validated.txt
```

Verified on Python 3.14.0 with pandas 3.0.1. The pinned file records all 143 packages present in the validated environment. The pin for `pywinpty`, a Windows-only terminal library pulled in by the Jupyter tooling, carries the environment marker `sys_platform == "win32"`, so the same file installs on Linux and macOS.

D.2 Test suite

```
python -m pytest -q
```

Expected: 43 passed.

D.3 The authoritative robustness round

Regenerates its datasets from recorded seeds and evaluates every condition, producing the figures in sections 6.4, 6.5 and 6.6 and in Appendix B.

```
python scripts/run_robustness_validation.py
```

D.4 Component breakdown

Recomputes the component contribution in section 6.3 from the recorded predictions rather than from a fresh run.

```
python scripts/component_breakdown.py
```

D.5 Feature validity and effect sizes

Produces the tables in section 6.2 and Appendix C.

```
python scripts/feature_validation.py --data-paths
data/synthetic_datasets/n500/audit_logs_surveycto.csv --ground-truth
data/synthetic_datasets/n500/ground_truth_labels.csv
```

The ODK comparison in section 6.2 uses the same command with `audit_logs_odk.csv` and `--output-dir output/feature_validation_odk`. Under a different pandas version the regenerated values agree with the recorded ones to within floating-point rounding rather than byte for byte.

D.6 Full workflow

Generates its own labelled validation bundle, including the canonical suite under `data/workflow_validation/canonical_validation/` that D.10 evaluates, and does not overwrite user-supplied files under `data/`. It must run before D.10. It also evaluates the n10000 pack, which is not committed, so the pack is generated from its recorded seed first; a clean run without that step fails at the n10000 evaluation.

```
python tests/generate_benchmark_datasets.py --dataset-size 10000 --seed 42 --
output-dir data/synthetic_datasets/n10000
```

```
python scripts/run_complete_workflow.py
```

D.7 Scale check

Uses the n10000 pack generated in D.6.

```
python scripts/analyze_pipeline_metrics.py --datasets n500 n2000 n10000 --formats surveycto --output-dir output
```

D.8 Strict examiner audit

The end-to-end audit and the source of the runtime figures in section 6.7. It checks raw-format consistency against the format guide, normalisation into the intermediate representation, manual recalculation of selected features, anomaly score reconstruction, reproducibility of precision and recall from ground truth, fabricated against normal feature separation, baseline comparisons, cross-format parity, and scale behaviour between n500 and n10000.

```
python scripts/strict_examiner_audit.py
```

D.9 Report figures

Regenerates Figures 1 and 2 from values transcribed from the recorded output files.

```
python scripts/build_report_figures.py
```

D.10 Canonical validation set

The contract test behind section 6.1. Requires the bundle generated by D.6.

```
python run_pipeline.py --inputs data/workflow_validation/canonical_validation/odk.csv data/workflow_validation/canonical_validation/surveycto.csv --parser auto --output-dir output --label canonical_validation --ground-truth data/workflow_validation/canonical_validation/ground_truth_labels.csv
```

Expected: precision 1.0000, recall 1.0000, F1 1.0000, with the four speeding interviews categorised suspicious rather than anomalous (section 6.1).

D.11 Word and PDF renderings of this report

Rebuilds the Word deliverables from their markdown sources, then measures the page numbers in the contents and the lists of tables and figures by exporting the document and reading the

pages back out of it. The second command requires Microsoft Word and writes both the `.docx` and the `.pdf`.

```
python scripts/build_docx_deliverables.py
python scripts/paginate_report.py
```

D.12 Baselines on every run of the robustness round

Produces Table 6a from the recorded features of D.3, and reproduces the two baseline rows of Table 3 as the `benchmark_n500_surveycto` rows.

```
python scripts/baseline_breakdown.py
```

D.13 Platform parity on matched histories

Regenerates the canonical n500 benchmark and both stress datasets with one history written into both formats, and produces Table 6b.

```
python scripts/run_platform_parity.py
```

D.14 Peak memory

Produces Table 8a. The n10000 size requires the pack generated in D.6, and is measured with `--datasets n10000 --formats surveycto --output-dir output/memory_n10000`.

```
python scripts/measure_memory.py --datasets n500 n2000 --formats surveycto odk
```

Appendix E. Reference verification record

Every reference was checked against its primary source for authors, title, venue, year and pages, and the DOI or URL used is given. The last column records how far the supporting passage has been verified, using three levels: **full text** means the source was obtained and read at the passage cited; **abstract** means only the abstract or publisher record was consulted, so the claim is held at the level that record supports; **not consulted** means the work has not been obtained, and the report cites it only for its scope, never for a specific finding. A reference that fails the bibliographic check is removed from the report, not annotated.

The bibliographic checks and the full-text reading recorded here were carried out with an AI tool on 10 and 16 September 2026, as episode 11 of Appendix A records. Six sources remain at abstract or not-consulted level; obtaining them from the university library and completing the last column is the outstanding work on this appendix, and it is listed at the end.

Table E1. Verification of every reference, including those withdrawn.

Reference	DOI or URL	Bibliographic check	Passage verification
Wenz, Kreuter and Janke (2019), *Survey Research Methods* 13(1), pp. 35-47	https://ojs.ub.uni-konstanz.de/srm/issue/view/140	**Failed.** The issue holds seven articles; pp. 23-44 are Jäckle et al. and pp. 45-71 are Read. No article by these authors, and none on fabrication, appears in it. **Withdrawn.**	Not applicable
Conrad, Tourangeau and Couper	none found	**Failed.** Not found in the journal or	Not applicable

Reference	DOI or URL	Bibliographic check	Passage verification
(2013), 'Adaptive survey design and paradata', *Journal of Official Statistics* 29(2)		in searches of the authors' published work. **Withdrawn.**	
Chawla et al. (2002), *Journal of Artificial Intelligence Research* 16, pp. 321-357	https://doi.org/10.1613/jair.953	Details correct, but the paper describes SMOTE, an oversampling method, and does not support the claims it was cited for. **Withdrawn.**	Full text: does not support the claim
Benjamini and Hochberg (1995), *JRSS Series B*	https://doi.org/10.1111/j.2517-6161.1995.tb02031.x	Correct	Abstract; cited only as the alternative procedure to

Reference	DOI or URL	Bibliographic check	Passage verification
57(1), pp. 289-300			Bonferroni
Birnbaum et al. (2013), *CHI '13*, pp. 2911-2920	https://doi.org/10.1145/2470654.2481404	Correct	Full text; accuracy figures p. 2911, feature construction p. 2913
Bredl, Winker and Kötschau (2012), *Survey Methodology* 38(1), pp. 1-10	https://www150.statcan.gc.ca/n1/pub/12-001-x/2012001/article/11680-eng.pdf	Correct	Full text; duration indicator p. 2
Bushery et al. (1999), ASA Survey Research Methods Section Proceedings, pp. 316-320	http://www.asasrms.org/Proceedings/papers/1999_053.pdf	Correct	Full text; both quotations p. 317
Chandola, Banerjee and Kumar	https://doi.org/10.1145/1541880.1541882	Correct	Full text; univariate definition

Reference	DOI or URL	Bibliographic check	Passage verification
(2009), *ACM Computing Surveys* 41(3), article 15			§2.1, unsupervised assumption §3. The paper does not discuss masking, and the earlier citation for it has been corrected
Cohen (1988), *Statistical Power Analysis for the Behavioral Sciences*, 2nd edn	https://doi.org/10.4324/9780203771587	Correct. Confirmed in the Crossref registry; the DOI identifies Routledge's 2013 reissue of the same second edition	Not consulted; cited only for the conventional effect-size measure
Conrad, Tourangeau,	https://ojs.ub.uni-konstanz.de/srm/article/view/6304	Correct	Full text; absolute

Reference	DOI or URL	Bibliographic check	Passage verification
Couper and Zhang (2017), *Survey Research Methods* 11(1), pp. 45-61			definition of speeding and the 350 ms per word benchmark, section 1
Couper (1998), ASA Survey Research Methods Section Proceedings, pp. 41-49	http://www.asasrms.org/Proceedings/papers/1998_006.pdf	Pages were missing in the first submission; pp. 41-49 added	Full text; process data p. 44, keystroke files p. 45
Finn and Ranchhod (2017), *World Bank Economic Review* 31(1), pp. 129-157	https://doi.org/10.1093/wber/lhv054	Correct. Confirmed in the Crossref registry, which records online publication in 2015 ahead of the 2017	Full text; prevalence, detection during fieldwork and the panel-estimator finding, p. 129. The first submission's

Reference	DOI or URL	Bibliographic check	Passage verification
		issue	wording, that the fabrication affected "some key covariates", was not the authors' and has been replaced with their own finding
Goel, Abraham and Lahoti (2022), IZA Discussion Paper No. 15041	https://docs.iza.org/dp15041.pdf	Correct	Full text; flags and dynamic benchmarking p. 3 and p. 5, the caution about evidence of wrongdoing p. 11. The published version, Goel and Abraham (2025) in

Reference	DOI or URL	Bibliographic check	Passage verification
			Development Policy Review, has not been consulted, so the working paper is cited
Groves et al. (2009), *Survey Methodology*, 2nd edn	ISBN 978-0-470-46546-2	Correct	Not consulted; cited only for the total survey error framework at the level of the book's scope
Hartung et al. (2010), *ICTD '10*, pp. 1-12	https://doi.org/10.1145/2369220.2369236	Correct; page range confirmed in the Crossref registry	Full text; description of the suite p. 1
Hawkins (1980), *Identification*	https://doi.org/10.1007/978-94-015-3994-4	Correct. Confirmed in the	Not consulted; no longer

Reference	DOI or URL	Bibliographic check	Passage verification
of Outliers		Crossref registry, which lists Springer's digitised edition of the 1980 Chapman and Hall book	cited for a specific claim
Hodge and Austin (2004), *Artificial Intelligence Review* 22, pp. 85-126	https://doi.org/10.1007/s10462-004-4304-y	Correct	Full text; single-dimensional methods and boxplots, section 3
Hoogeveen and Pape (eds) (2020), *Data Collection in Fragile States*	https://doi.org/10.1007/978-3-030-25120-8	Correct; open access, first published online October 2019	Full text of the foreword, from which the quotation is taken; individual chapters not read
Hubert and	https://doi.org/10.1016/j.csda.2007.11.008	Correct	Full text of

Reference	DOI or URL	Bibliographic check	Passage verification
Vandervieren (2008), *Computational Statistics & Data Analysis* 52, pp. 5186-5201			the authors' accepted manuscript; exceedance probabilities, the practitioner quotation and the semi-interquartile fence all §1. Section numbers are cited rather than pages because the manuscript is not paginated as published
Iglewicz and Hoaglin (1993), *How to Detect and Handle Outliers*	ISBN 978-0-87389-247-6	Correct; volume 16 of the ASQC Basic References in Quality	Not consulted; cited only for the unsuitability of normality-

Reference	DOI or URL	Bibliographic check	Passage verification
		Control	based methods on skewed data
Japkowicz and Stephen (2002), *Intelligent Data Analysis* 6(5), pp. 429-449	https://doi.org/10.3233/IDA-2002-6504	Correct	Abstract; the prevalence-sensitivity claim must be confirmed against the full text
Kerby (2014), *Comprehensive Psychology* 3	https://doi.org/10.2466/11.IT.3.1	Correct	Abstract, which states the simple difference formula used here
Kreuter (ed.) (2013), *Improving Surveys with Paradata*	https://doi.org/10.1002/9781118596869	Correct	Not consulted; cited only for the scope of the volume. The two specific findings attributed to it in the first

Reference	DOI or URL	Bibliographic check	Passage verification
			submission have been removed
Krosnick (1991), *Applied Cognitive Psychology* 5(3), pp. 213-236	https://doi.org/10.1002/acp.2350050305	Correct; confirmed in the Crossref registry	Abstract. The satisficing account attributed to it is also stated, and attributed to this paper, in Krosnick (1999, pp. 39, 41), which was read at full text
Krosnick (1999), 'Maximizing questionnaire quality', in Robinson, Shaver and Wrightsman (eds), *Measures of*	Academic Press chapter; no DOI	Correct; the chapter's title page gives the book, editors, publisher and first page	Full text; satisficing p. 39 and p. 41, fatigue p. 40

Reference	DOI or URL	Bibliographic check	Passage verification
Political Attitudes, pp. 37-57			
McKinney (2017), *Python for Data Analysis*, 2nd edn	ISBN 978-1-491-95766-0	Correct	Not consulted; cited as the reference for the software stack
ODK (n.d.), *Form audit log*	https://docs.getodk.org/form-audit-log/	Correct	Full page read: columns event, node, start, end; epoch-millisecond timestamps; the event list
Schwanhäuser, Sakshaug and Kosyakova (2022), *Public Opinion Quarterly*	https://pmc.ncbi.nlm.nih.gov/articles/PMC8944307/	Correct	Full text; indicators, comparison of methods and the lower-variance finding.

Reference	DOI or URL	Bibliographic check	Passage verification
86(1), pp. 51-81			Claims are paraphrased rather than quoted
SurveyCTO (n.d.-a), *Field type: text audit*	https://docs.surveycto.com/02-designing-forms/01-core-concepts/03zd.field-types-text-audit.html	Correct	Full page read: the default and eventlog layouts
SurveyCTO (n.d.-b), *Collecting high-quality data*	https://docs.surveycto.com/04-monitoring-and-management/02-managing-for-quality/01.collecting-high-quality-data.html	Correct	Full page read: speed limits, speed violation counts, automated quality checks
Tukey (1977), *Exploratory Data Analysis*	book, no DOI	Correct	Not consulted; the fence construction is cited as described by Hubert and Vanderviere n (2008, §1)
Yan and	https://doi.org/10.1002/acp.1331	Correct;	Full text;

Reference	DOI or URL	Bibliographic check	Passage verification
Tourangeau (2008), *Applied Cognitive Psychology* 22(1), pp. 51-68		confirmed in the Crossref registry	determinants of response time p. 51, positive skew p. 52, the disclaimer about data quality p. 56. The draft's claim that the paper links fast answers to lower quality was not supported, since the authors say their emphasis is not on that relation, and has been corrected
Zhang and	https://ojs.ub.uni-	Correct	Full text;

Reference	DOI or URL	Bibliographic check	Passage verification
Conrad (2014), *Survey Research Methods* 8(2), pp. 127-135	konstanz.de/srm/article/view/5453		the 300 ms per word threshold and the straightlining association, section 2

Outstanding. Every reference was confirmed to exist through the Crossref DOI registry, the Open Library ISBN record, or the publisher's own document. Ten are cited from their abstract or publisher record rather than their full text: Benjamini and Hochberg (1995), Cohen (1988), Groves et al. (2009), Hawkins (1980), Iglewicz and Hoaglin (1993), Japkowicz and Stephen (2002), Kerby (2014), Kreuter (2013), McKinney (2017) and Tukey (1977). Each is cited only for what its abstract or record supports, and none carries a load-bearing result. Krosnick (1991) is also at abstract level, but the claim attributed to it is independently supported by Krosnick (1999), read at full text.

Table E2. Sources checked but not cited. These were verified during the rebuild of section 2 and remain available if the themes are extended.

Candidate	DOI or URL	Bibliographic check
Choumert-Nkolo, Cust and Taylor (2019), *Review of Development Economics* 23(2), pp. 598-618	https://ideas.repec.org/a/bla/rdevec/v23y2019i2p598-618.html	Correct; full text not obtained
Couper and Kreuter (2013), *JRSS Series A*	https://doi.org/10.1111/j.1467-985X.2012.01041.x	Correct; full text not

Candidate	DOI or URL	Bibliographic check
176(1), pp. 271-286		obtained
Groves and Lyberg (2010), *Public Opinion Quarterly* 74(5), pp. 849-879	https://doi.org/10.1093/poq/nfq065	Correct; full text not obtained
Kuriakose and Robbins (2016), *Statistical Journal of the IAOS* 32, pp. 283-291	https://doi.org/10.3233/SJI-160978	Correct; concerns duplication rather than timing
Olbrich, Kosyakova, Sakshaug and Schwanhäuser (2024), *Journal of Survey Statistics and Methodology* 12(1), pp. 14-35	https://doi.org/10.1093/jssam/smac036	Correct; full text not obtained

Appendix F. Feature-to-source map

Every feature, the source the first submission gave for it in `docs/feature_documentation.md`, and the source it rests on after the rebuild of section 2. Seventeen features had been attributed to one of the two references that cannot be located. Where the rebuilt literature supports a feature, the replacement source is named with its page; where it does not, the feature is described as a design choice, with whatever partial support exists stated rather than implied.

Counting the entries below, 23 of the forty features are attributed to a named source and 17 are design choices. That is a fair summary of the position: the timing, variability and pace-trajectory features are grounded in the falsification and response time literature, while the navigation, pause

and completion features are this project's own constructions. Section 8 treats the second group as a limitation, and the evaluation in section 6.2 is silent on most of them in any case.

Table F1. Source for each of the forty features.

ID	Feature	Source in the first submission	Source after revision
Q-01	Response time	Yan and Tourangeau (2008)	Satisficing as shortcut cognitive effort (2.2): Krosnick (1999, p. 39); determinants of response time: Yan and Tourangeau (2008, p. 51); Birnbaum et al. (2013, p. 2913), whose instance-level features aggregate per-question answer times
Q-02	Time to first interaction	Couper (1998)	Birnbaum et al. (2013, p. 2913), whose instance-level features aggregate per-question answer times, including delay to first edit
Q-03	Edit count	Conrad et al. (2013), withdrawn	Couper (1998, p. 45), for the keystroke record from which edits are counted; Birnbaum et al. (2013, p. 2913), whose instance-level features aggregate per-question answer times
Q-04	Revisit flag	Wenz et al. (2019), withdrawn	Design choice. Nearest support is Birnbaum et al. (2013, p. 2913), whose conditional-path features test whether fabricators take short routes through an instrument
Q-05	Revisit count	Kreuter (2013)	Design choice. Nearest support is Birnbaum et al. (2013, p. 2913), whose conditional-path features test whether fabricators take short routes through an instrument
Q-06	Field completion ratio	Groves et al. (2009)	Design choice. A data-presence proxy imposed by the log (3.2)

ID	Feature	Source in the first submission	Source after revision
Q-07	Idle time	Conrad et al. (2013), withdrawn	Design choice. The disengagement profile it serves has no verified source (5.1)
Q-08	Keystroke count (event-count proxy)	Yan and Tourangeau (2008)	Couper (1998, p. 45): the audit trail is a record of interviewer input, of which the event count is the available proxy
Q-09	Constraint trigger count	Couper (1998)	Couper (1998, p. 45), for the trace of interviewer input; the diagnostic use is a design choice
Q-10	First-to-last interaction span	Wenz et al. (2019), withdrawn	Birnbaum et al. (2013, p. 2913), whose instance-level features aggregate per-question answer times, whose total-time feature has the same construction
S-01	Mean section pace	Kreuter (2013)	Bredl et al. (2012, p. 2), who treat time to complete a module as an indicator
S-02	Section duration	Groves et al. (2009)	Bredl et al. (2012, p. 2), as above
S-03	Within-section variance	Wenz et al. (2019), withdrawn	Lower variability in falsified interviews (2.3): Bredl et al. (2012, p. 2); Schwanhäuser et al. (2022)
S-04	Section-to-section pace ratio	Conrad et al. (2013), withdrawn	Design choice. It is also constant by construction on equal-sized sections (3.2)
S-05	Pause count over 5 s	Yan and Tourangeau (2008)	Design choice. The disengagement profile it serves has no verified source (5.1)

ID	Feature	Source in the first submission	Source after revision
S-06	Pause count over 30 s	Kreuter (2013)	Design choice. The disengagement profile it serves has no verified source (5.1)
S-07	Mean pause duration	Conrad et al. (2013), withdrawn	Design choice. The disengagement profile it serves has no verified source (5.1)
S-08	Maximum pause duration	Wenz et al. (2019), withdrawn	Design choice. The disengagement profile it serves has no verified source (5.1)
S-09	Section back-navigation count	Couper (1998)	Design choice. Nearest support is Birnbaum et al. (2013, p. 2913), whose conditional-path features test whether fabricators take short routes through an instrument
S-10	Section completion rate	Groves et al. (2009)	Design choice. A data-presence proxy imposed by the log (3.2)
V-01	Total interview duration	Yan and Tourangeau (2008)	Falsification timing signature (2.3): Bushery et al. (1999, p. 317); Schwanhäuser et al. (2022)
V-02	Early pace	Kreuter (2013)	Yan and Tourangeau (2008, p. 51), who find response time depends on a question's position within the questionnaire; Krosnick (1999, p. 40), on responses degrading after respondents "have become fatigued answering a lot of prior questions"
V-03	Late pace	Conrad et al. (2013), withdrawn	As V-02

ID	Feature	Source in the first submission	Source after revision
V-04	Pace trajectory ratio	Wenz et al. (2019), withdrawn	As V-02
V-05	Back-navigation rate	Couper (1998)	Design choice. Nearest support is Birnbaum et al. (2013, p. 2913), whose conditional-path features test whether fabricators take short routes through an instrument
V-06	Back-navigation total	Groves et al. (2009)	Design choice. Nearest support is Birnbaum et al. (2013, p. 2913), whose conditional-path features test whether fabricators take short routes through an instrument
V-07	Start hour	Wenz et al. (2019), withdrawn	Design choice. Bushery et al. (1999) use date and time stamps for interviewer screening, but not the hour of day as such
V-08	Day of week	Kreuter (2013)	Design choice, as V-07
V-09	Duration relative to the sample median	Yan and Tourangeau (2008)	Schwanhäuser et al. (2022), who use relative as well as absolute interview duration; Goel et al. (2022, p. 5) for benchmarking against comparable peers
V-10	Run of 20 or more events without back-navigation	Conrad et al. (2013), withdrawn	Design choice. Nearest support is Birnbaum et al. (2013, p. 2913), whose conditional-path features test whether fabricators take short routes through an instrument
T-01	Response time variance	Wenz et al. (2019),	Lower variability in falsified interviews (2.3): Bredl et al. (2012, p. 2); Schwanhäuser et al.

ID	Feature	Source in the first submission	Source after revision
		withdrawn	(2022)
T-02	Response time IQR	Kreuter (2013)	Lower variability in falsified interviews (2.3): Bredl et al. (2012, p. 2); Schwanhäuser et al. (2022)
T-03	Response time entropy	Conrad et al. (2013), withdrawn	Lower variability in falsified interviews (2.3): Bredl et al. (2012, p. 2); Schwanhäuser et al. (2022). The entropy formulation itself is a design choice (3.3)
T-04	Response time skewness	Yan and Tourangeau (2008)	Yan and Tourangeau (2008, p. 52), who report that response time distributions are "asymmetrical and positively skewed"
T-05	Response time excess kurtosis	Wenz et al. (2019), withdrawn	Lower variability in falsified interviews (2.3): Bredl et al. (2012, p. 2); Schwanhäuser et al. (2022)
T-06	Response time mean	Groves et al. (2009)	Birnbaum et al. (2013, p. 2913), whose instance-level features aggregate per-question answer times
T-07	Response time median	Yan and Tourangeau (2008)	Birnbaum et al. (2013, p. 2913), whose instance-level features aggregate per-question answer times, who use median answer time
T-08	Response time coefficient of variation	Wenz et al. (2019), withdrawn	Lower variability in falsified interviews (2.3): Bredl et al. (2012, p. 2); Schwanhäuser et al. (2022)
T-09	Response time minimum	Couper (1998)	Birnbaum et al. (2013, p. 2913), whose instance-level features aggregate per-question answer times, who use minimum answer time

ID	Feature	Source in the first submission	Source after revision
T-10	Response time maximum	Conrad et al. (2013), withdrawn	Design choice, as the counterpart of T-09